

对称性考虑（课外扩展）

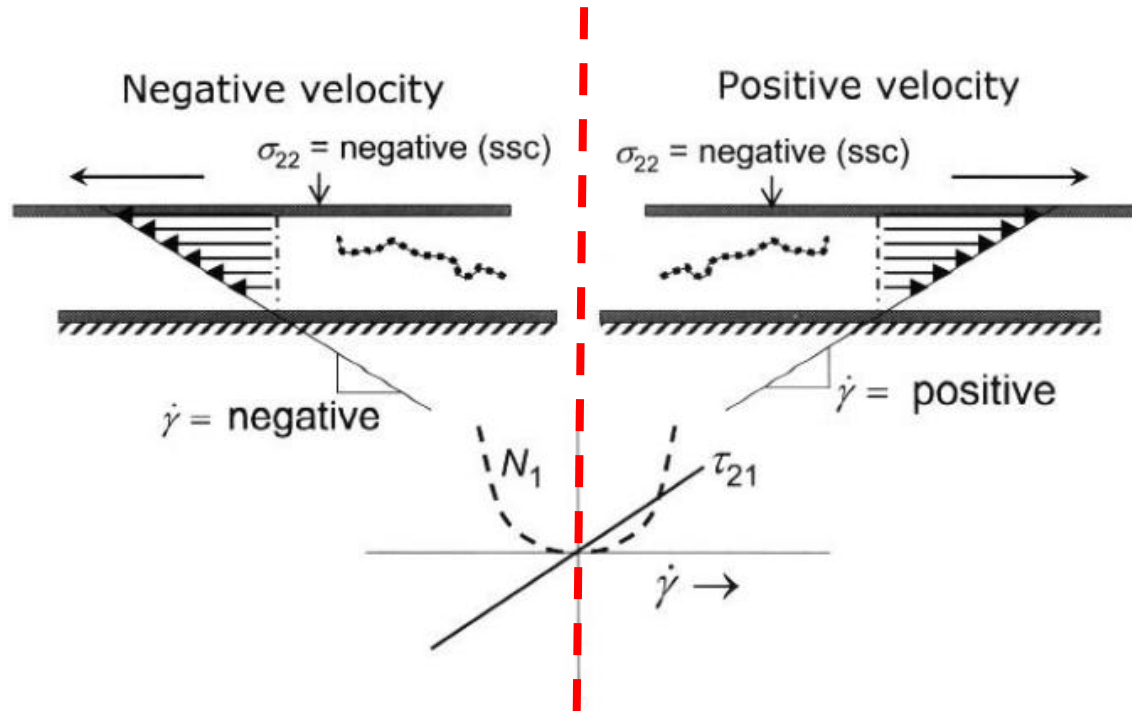


Figure 6-2. With the positive direction established as left to right, the shear stress becomes negative when the plate is moved steadily from right to left (left panel). The first normal-stress difference, however, retains its sign. Thus we expect N_1 to depend on $\dot{\gamma}^2$ at low shear rates. The dotted line depicts a polymer molecule in the flow field; it is stretched in the flow direction regardless of the sign of the velocity gradient.

左边和右边两种情况是镜面（竖直虚线所在的垂直于纸面的平面）对称的，所以法向应力差只能和剪切速率的偶数次方成正比

总结：聚合物熔体流动中的弹性行为

掌握三个问题：

1) 聚合物弹性行为的宏观表现形式和定量表征：

入口压力降、爬杆、挤出胀大、不稳定流动（鲨鱼皮症、熔体破裂）

2) 聚合物弹性行为的本质：

力学原因：第一法向应力差；微观原因：外力作用下分子链被拉伸，偏离无应力时的平衡构象，用可逆弹性应变定量表征。

3) 弹性强弱的判据：韦森堡数 $N_w = \lambda \dot{\gamma}$

4) 聚合物弹性行为的利弊以及如何削弱不利的弹性行为？

利：加工稳定性（拉伸硬化）、取向导致力学性能提高；

弊：加工不稳定、表面粗糙

措施：降低可逆应变，降低韦森堡数

第7节 影响聚合物流变行为的主要因素

A. 影响熔体粘度 η 的因素可以分为三大类：

1. 加工条件（温度、剪切速率、压力等）
2. 分子结构特征（分子量及其分布、支链、分子间作用）
3. 配方因素（添加剂：润滑剂、填料等）

B. 影响粘度 η 的微观因素：

1. 自由体积（本体中）
2. 流体力学体积（稀溶液中）
3. 取向（较大的应力下）
4. 聚合物单元间的摩擦系数（取决于分子间相互作用等因素）

7.1 温度对粘度的影响（用活化能或自由体积解释）

1. 在有限的温度范围内， $T > T_g + 100^\circ\text{C}$
聚合物粘度， $\eta \sim T$ 关系, Andrade 公式: $\eta = Ae^{\frac{E_\eta}{kT}}$

E_η 是粘流活化能，是分子链流动时用于克服分子间作用力以便更换位置所需要的能量。

2. 很宽温度范围， $T_g \sim T_g + 100^\circ\text{C}$ ，粘流活化能不再是常数，协同跃迁单元数随温度下降而增加。

WLF方程（以 T_g 为基准温度）：

$$\log \left[\frac{\eta(T)}{\eta(T_g)} \right] = - \frac{17.44(T - T_g)}{51.6 + (T - T_g)}$$

WLF方程适用范围：

1. $T_g \sim T_g + 100^\circ\text{C}$

2. 非晶态聚合物（如果是结晶聚合物，熔点以上才能熔融加工）

△实际加工中，要考虑加热时间（热降解）对粘度的影响，

熔融加工温度范围：粘流温度（ T_f ）—热分解温度（ T_d ）

表 一些高分子材料体系的粘流活化能

聚合物	$E_{\eta} /$ $\text{kcal} \cdot \text{mol}^{-1}$	E_{η} $/ \text{kJ} \cdot \text{mol}^{-1}$	聚合物	E_{η} $/ \text{kcal} \cdot \text{mol}^{-1}$	E_{η} $/ \text{kJ} \cdot \text{mol}^{-1}$
天然橡胶	0.25	1.04	PS	22~23	92~96
顺丁橡胶	2.3	9.6	PC	26~30	108.3~125
丁苯橡胶	3.1	12.9	PVC	35~40	147~168
丁腈橡胶	5.42	22.6			
聚二甲基硅 氧烷	4.0	16.7	醋酸纤维素	70	293.3
HDPE	6.3~7.0	26.3~29.2	ABS (20%橡胶)	26	108.3
LDPE	10~12.8	41.9~53.6	ABS (30%橡胶)	24	100
PP	10~11	41.9~46	ABS (40%橡胶)	21	87.5
PP (长支链较 多)	11~17	46~71.2			

7.2 剪切速率对粘度的影响

1. 一般剪切速率升高，粘度下降

但加工中要注意，不要超过临界剪切速率，熔体破裂

2. 粘度对剪切速率的敏感性指标： $\eta(100\text{S}^{-1}) / \eta(1000\text{S}^{-1})$
工业测试，熔融指数MFR(2.16Kg) / MFR(21.6Kg)，相同测试温度

3. 哪些高聚物对剪切速率敏感：

敏感的聚合物（这些聚合物有何特点？柔性分子链（静态柔性），外力作用下分子链易变形）：

聚苯乙烯、聚乙烯、聚丙烯、聚氯乙烯等

不敏感的聚合物：

聚酰胺、聚对苯二甲酸乙二酯等

7.3 压力对粘度的影响（用自由体积解释）

a. **P**升高，自由体积下降， η 增大

$$\eta_p = \eta_0 \exp(\alpha_p \cdot p)$$

where

η_p = viscosity at pressure p and constant shear stress τ_0

η_0 = viscosity at constant shear stress τ_0

α_p = pressure coefficient

b. 压力—温度等效性：**P**升高，相当于**T**下降

在维持 η 恒定的条件下，**每升高1个大气压，
相当于 温度降低 $(3\sim9) \times 10^{-2} \text{ }^\circ\text{C}$**

**塑料注射成型，熔体压力可能超过100MPa，
相当于温度下降了多少摄氏度？**

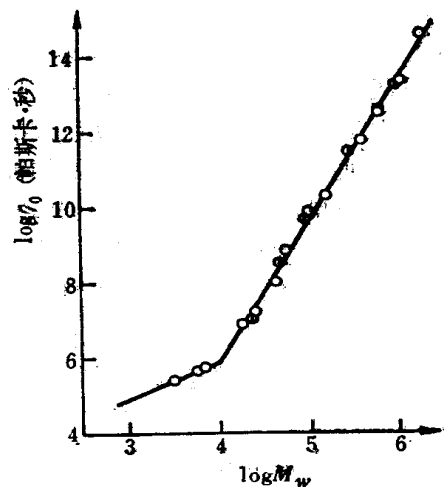
7.4 聚合物分子结构因素对 η 的影响

7.4.1 分子量和分子量分布的影响

7.4.2 支化结构的影响（支化、星形等）

7.4.1 分子量和分子量分布的影响

A. 聚合物分子量对零切粘度的影响



顺式聚异戊二烯的粘度对分子量关系

$$\eta_0 = K_1 \overline{M}_w \quad (\overline{M}_w < M_c)$$

$$\eta_0 = K_2 \overline{M}_w^m \quad (\overline{M}_w > M_c) \quad m = 3.1 - 3.4$$

M_c : 临界分子量, 发生分子链缠结的最小分子量

表 5-9 几种聚合物的临界分子量值

聚 合 物	M_c	聚 合 物	M_c
聚乙烯	4 000	天然橡胶	5 000
聚丙烯	7 000	聚丁二烯	5 900
聚碳酸酯	4 800	聚氯乙烯	11 000
聚氧乙烯	4 400	聚异丁烯	17 000
聚氧丙烯	5 800	聚甲基丙烯酸甲酯	18 400
聚乙烯醇	7 500	聚醋酸乙烯酯	24 500
尼龙 6	5 000	聚二甲基硅氧烷	24 500
尼龙 66	7 000	聚苯乙烯	31 200

B. 分子量对聚合物熔体粘度的影响

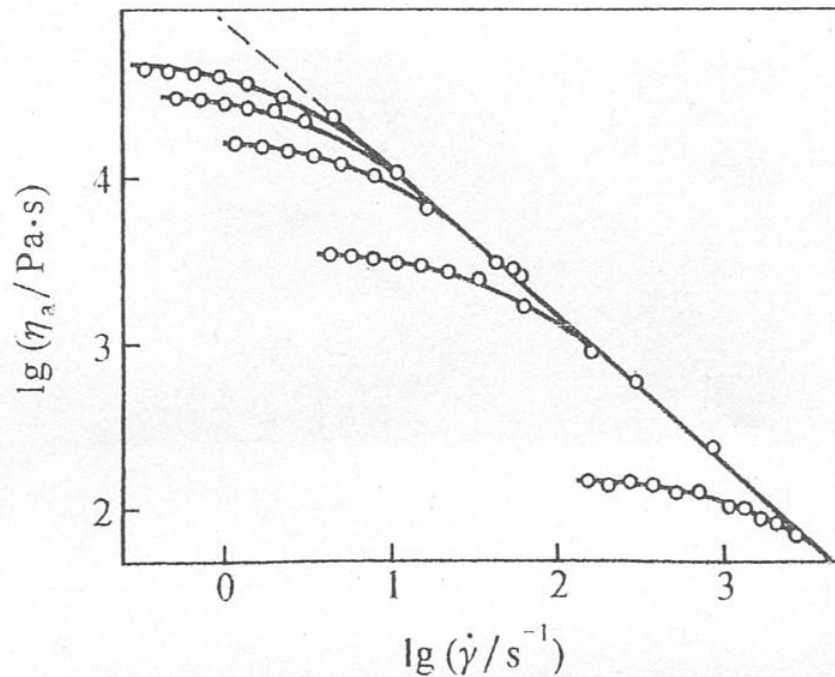
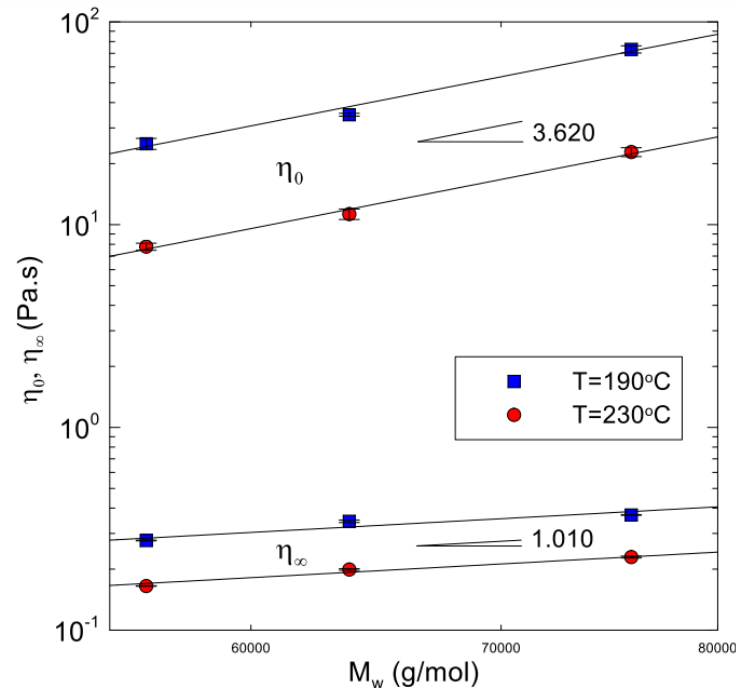


图6-10 183°C时几种不同分子量的聚苯乙烯的粘度与剪切速率的关系。从上到下各曲线的分子量分别是**242,000**；**217,000**；**179,000**；**117,000**；**48,500**

分子量显著影响零切粘度

重均分子量对零切粘度和无穷切粘度的影响



零切粘度随分子量变化很大；
高剪切速率下的粘度随分子量变化很小，
你推测是什么原因？

C. 分子量对粘弹性的影响

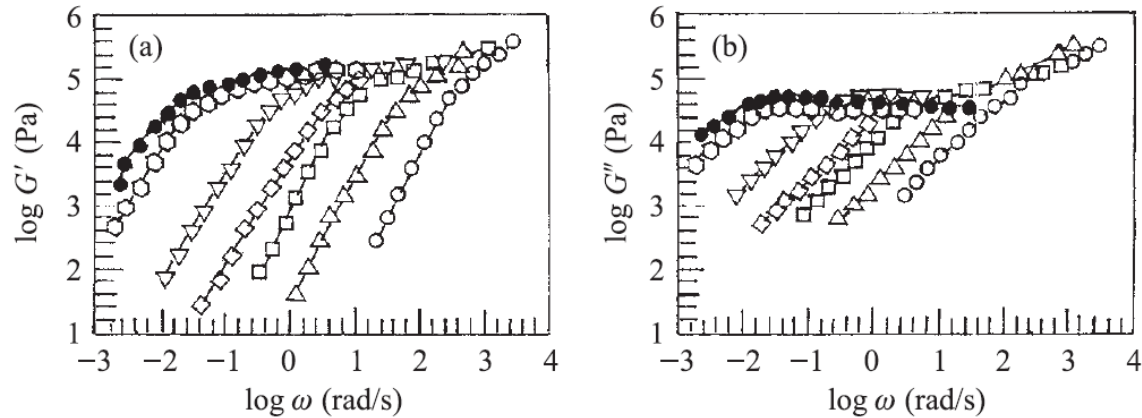


Figure 6.21 Plots of (a) $\log G'$ versus $\log \omega$ and (b) $\log G''$ versus $\log \omega$ at 167 °C for seven nearly monodisperse polystyrenes having different molecular weights: (○) L23, (△) L39, (□) L72, (◇) L89, (▽) L172, (◐) L315, and (●) L427. See Table 6.6 for information on the molecular weight of these polymers.

Table 6.6 Molecular weights of anionically synthesized polystyrenes

Sample Code	M_w	M_n	M_w/M_n	M_w/M_e
L23	23.4×10^3	21.8×10^3	1.08	1.3
L39	38.9×10^3	36.3×10^3	1.07	2.2
L72	72.4×10^3	68.3×10^3	1.06	4.0
L89	88.5×10^3	82.7×10^3	1.07	4.9
L172	172×10^3	161×10^3	1.07	9.6
L315	315×10^3	294×10^3	1.07	17.5
L427	427×10^3	407×10^3	1.07	23.7

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D. 熔体零切粘度和松弛时间的分子长度依赖性

$$\eta_0 = \lambda G$$

η_0 : 聚合物熔体的零切粘度

λ : 聚合物分子的松弛时间

G : 弹性模量

A. Rouse链: $\lambda \propto N^2$, $G \propto N^{-1}$, $\eta_0 = \lambda G \propto N^1$

B. 缠结高分子链: $\lambda = \frac{L_{tube}^2}{D} \propto \frac{N^2}{1/N} \propto N^3$ (reptation theory),
模量不随分子量改变 $G \propto N^0$, $\eta_0 = \lambda G \propto N^3$

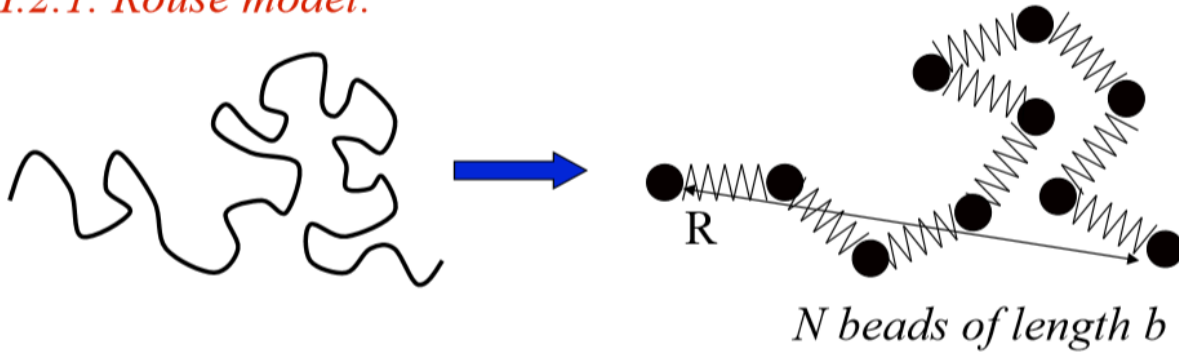
N : 一根分子链中的链段数目

L_{tube} : 缠结形成的管子的长度

D : 熔体分子链沿着缠结管的扩散系数

Rouse模型 (适用于高分子稀溶液或未缠结的高分子熔体)

1.2.1. Rouse model:



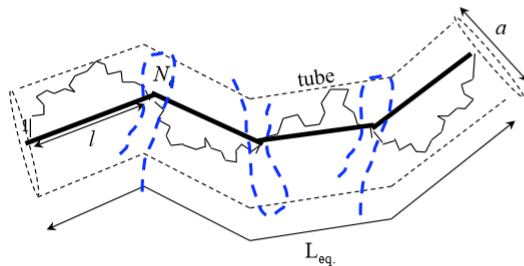
- Each bead has its own friction ζ

- Total friction: $\zeta_R = N\zeta \quad \longrightarrow \quad D_R = \frac{kT}{N\zeta}$

- Rouse time: $\tau_R \approx \frac{R^2}{D_R} \approx \frac{R^2}{kT / N\zeta} = \frac{\zeta}{kT} NR^2 \propto N^2$

Rouse链: 松弛时间 $\lambda \propto N^2$, $G \propto N^{-1}$, $\eta_0 = \lambda G \propto N^1$

Equilibrium state in a polymer melt: parameters



$$L_{eq} = Z \cdot l = \frac{N}{N_e} \cdot l$$

L_{eq} = length of the primitive path

Z = number of segments between two ent.

l = length of a segment between two ent.

a = tube diameter

b = length of a Khun segment

N_e = number of Khun segments between two entanglements

N = total number of Khun segments in the chain

Reptation

Curvilinear diffusion coefficient:

$$D_e = \frac{kT}{\zeta_{tot}} = \frac{kT}{N\zeta_0}$$

↑ Total friction of the chain
 ↑ Monomeric friction

Reptation time:

$$\tau_{rept} = \frac{L_{eq}^2}{D_e} \propto \frac{N^2}{N^{-1}} = N^3$$

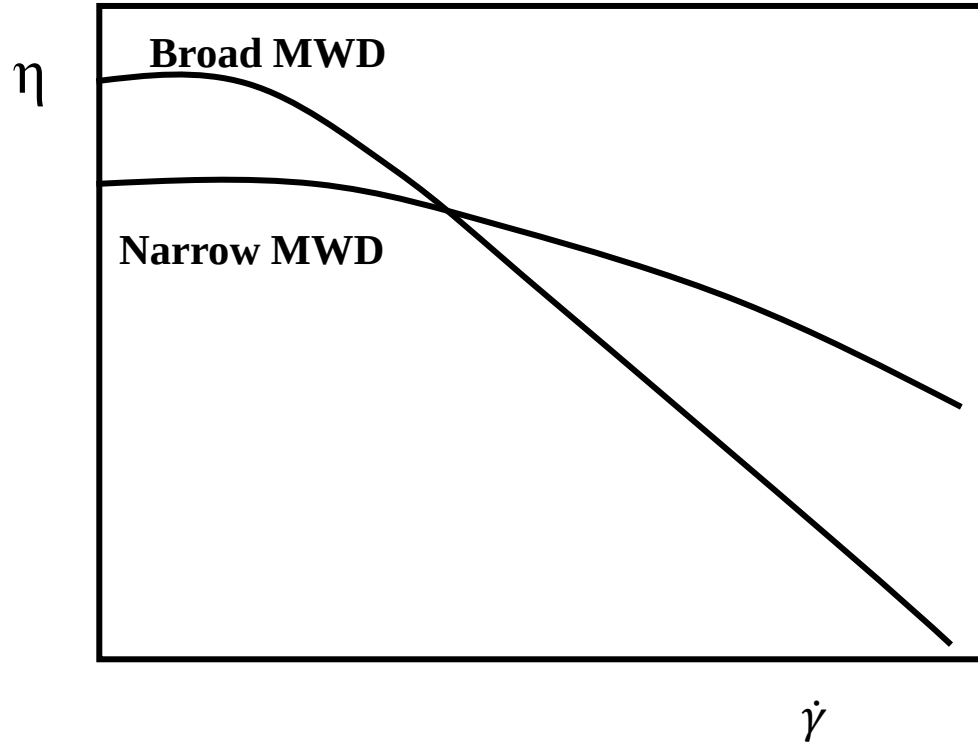
$$\tau_{Rept} \approx \tau_e \left(\frac{N}{N_e} \right)^3$$

• The molecular weight between two entanglements: M_e

• The plateau modulus:

$$G_N^0 = \frac{\rho N_A kT}{M_e} = \frac{\rho R T}{M_e} \quad (\times 4/5)$$

E. 聚合物分子量分布对 η 的影响



分布宽，非牛顿性显著， η 对剪切较敏感

低剪切速率的粘度和分子量3.4次方成正比，高分子量部分贡献更大
；中高剪切速率下，低分子量部分对粘度的贡献更大

F. 聚合物分子量分布对弹性的影响

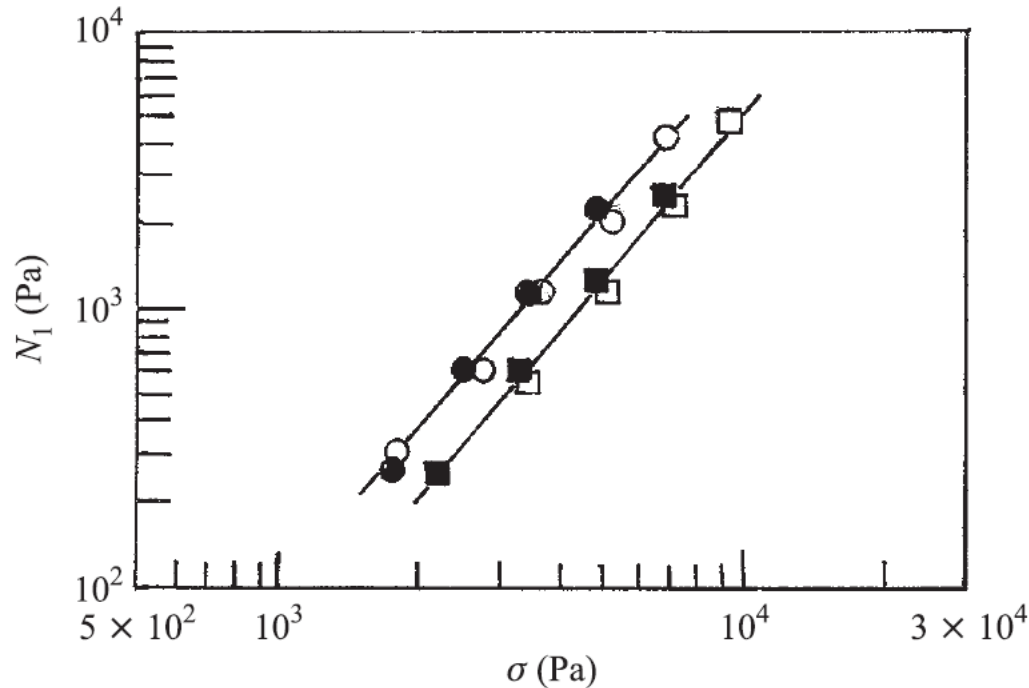


Figure 6.34 Plots of $\log N_1$ versus $\log \sigma$ for two commercial polystyrenes having different weight-average molecular weight (M_w) and polydispersity (M_w/M_n): (a) Styron 678 ($M_w = 2.29 \times 10^5$, $M_w/M_n = 4.31$) at 200 °C (○) and at 220 °C (●), and (b) Styron 685 ($M_w = 2.85 \times 10^5$, $M_w/M_n = 2.85$) at 200 °C (□) and at 220 °C (■).

1. The linear flexible homopolymer with **the higher polydispersity index is more elastic** than the homopolymer with the lower polydispersity index.
2. $\log N_1$ versus $\log \sigma$ plots are **independent of molecular weight for entangled, linear flexible homopolymers**.

重均分子量相近时，分子量分布宽，弹性更显著

G. 总结：分子量及其分布对加工性能的影响

1、从纯粹加工的角度来看，降低分子量肯定有利于改善材料的流动性。但分子量降低后必然影响材料的强度和弹性，因此需综合考虑。

能否由什么办法兼顾两者的优点？ 譬如耐压聚乙烯管PE100(周向耐压100bar, 10MPa),可以使用50年，既要有高强度又要有良好的加工性能。

2、不同的材料，因用途不同，加工方法各异，对分子量的要求不同。

3、塑料中，用于注射成型的树脂分子量应小些，用于挤出成型的树脂分子量可大些，用于吹塑成型的树脂分子量可适中。通常，熔融加工的剪切速率越高，要求的分子量越小（熔融指数越高）。譬如薄膜级PP，熔融指数0.5-2g/10min; 挤出级，MFR 2-8g/10min; 注射级，MFR 10-80（高流动性）；熔喷纤维级，MFR 1500g/10min。熔融指数（简称熔指）越大，流动性越好，分子量越低。

3、分子量分布宽，有利于挤出和注射加工（利用剪切变稀）；
分子量分布窄，有利于纺丝（利用拉伸稳定性）。

7.4.2 聚合物支化结构对 η 的影响

A. 链支化:

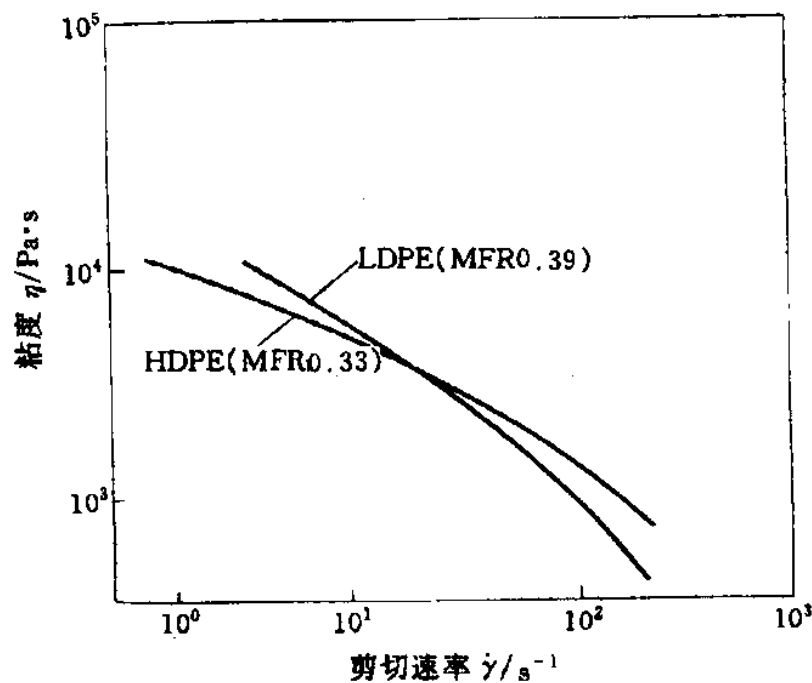
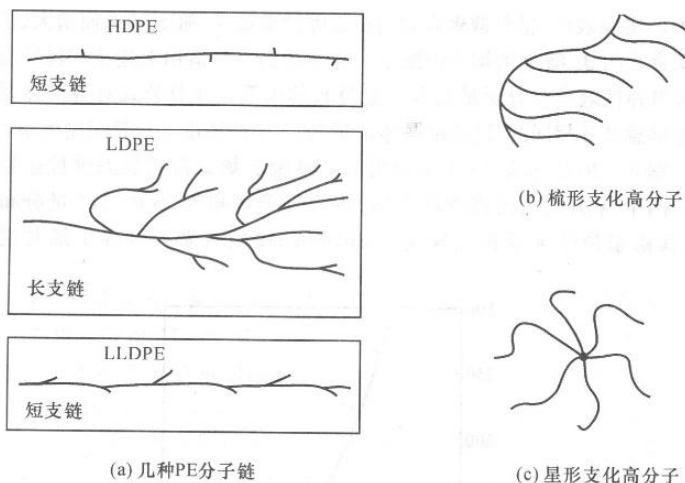
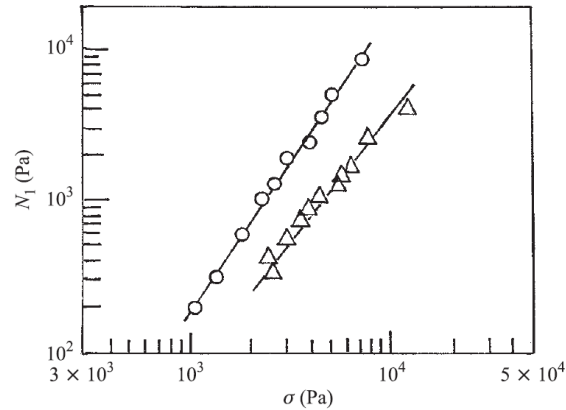


图 4-11 高、低密度聚乙烯的粘度与剪切速率的关系

低密度聚乙烯 (LDPE) 有长支化, 剪切变稀更显著
高密度聚乙烯 (HDPE) 有少量短支链, 剪切变稀不如 LDPE 显著

长支化显著提高第一法向应力差



长支化显著提高零切粘度和剪切变稀

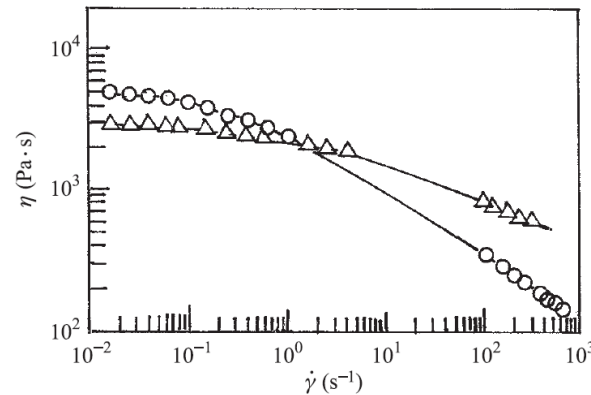


Figure 6.35 Plots of $\log \eta$ versus $\log \dot{\gamma}$ for an LDPE ($\circ$) and an LLDPE ($\triangle$) at 200°C . (Reprinted from Kwack and Han, *Journal of Applied Polymer Science* 28:3419. Copyright © 1983, with permission from John Wiley & Sons.)

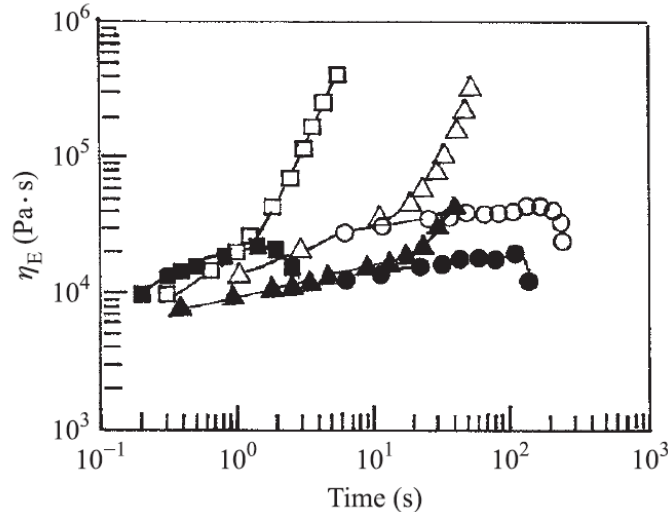


Figure 6.37 Time evolution of uniaxial elongational viscosity (η_E) at 200°C for an LDPE (open symbols) and an LLDPE (filled symbols) at various elongation rates (s^{-1}): ($\circ$, $\bullet$) 0.01, ($\triangle$, $\blacktriangle$) 0.1, and ($\square$, $\blacksquare$) 1.0.

长支化显著提高拉伸粘度



LDPE有长支链，LLDPE含短支链

B. 多臂星形高分子分子量对粘度的影响

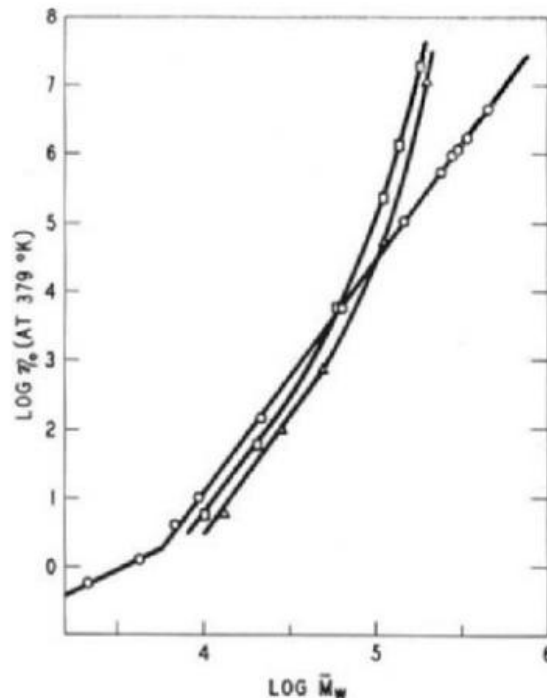


Figure 5.19 Zero-shear viscosity versus molecular weight for polybutadiene linears (circles), three-arm stars (squares), and four-arm stars (triangles). At low M_w , the data for the branched samples fall below the line for linear polymers, while at higher M_w they cross the line and become higher. From Kraus and Gruver [89].

多臂 (≥ 3) 星形高分子臂长超过缠结长度后，
缠结更多，粘度随分子量增加升高更显著

C. 超支化对 η 的影响

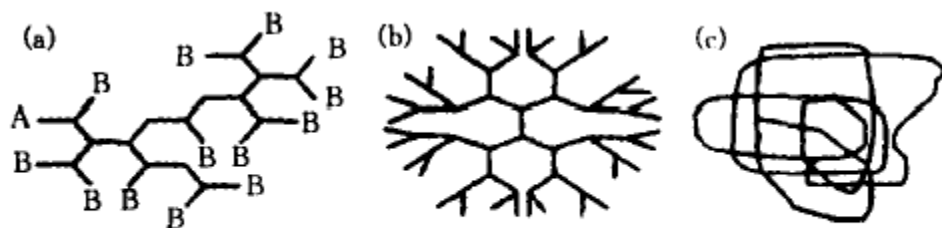


图1 超支化聚合物(a)、树枝状大分子(b)、
线型聚合物(c)的分子结构示意图

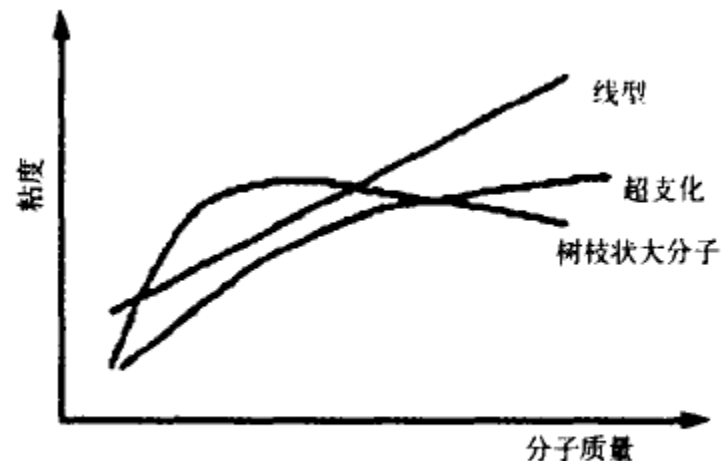


图2 3种聚合物粘度随分子质量变化示意图

树枝状大分子的粘度在分子量超过一定数值后，反而随分子量提高而下降。原因：分子间缠结减少。

D. 总结：支化的主要影响规律

(1) 一般说，短支链（梳型支化）对材料粘度的影响甚微。对高分子材料粘度影响大的是长支链（星型支化）的形态和长度。

(2) 若支链虽长，但其长度还不足以使支链本身发生缠结，这时体系自由体积增大。与分子量相当的线型聚合物比，支化聚合物的粘度要低些（ $\eta_b < \eta_l$ ）。

(3) 若支链相当长，长支链本身发生缠结，长支化聚合物的非牛顿性较强，在高剪切速率下，支化聚合物比分子量相当的线型聚合物的粘度低。

(4) 超支化大分子的粘度在分子量超过一定数值后，反而随分子量提高而下降。

(5) 长支链使熔体强度增加，有利于吹膜、发泡。

7.5 体系组成对 η 的影响

1. 加低分子物（溶剂、增塑剂、润滑剂），流动促进剂（如液晶，流动取向），降低粘度；

2. 共混（不同高分子）

共混是高分子改性三大方法之一（连续纤维增强、填充、共混）。

共混（blending）是指加入高分子和高分子混合，研究共混高聚物熔体的流变学，对于控制共混物的相结构和最终性能有重要指导意义。

7.5.1 高分子共混体系的流变行为

A. 高分子—高分子共混五项原则

①极性相匹配——从结构上看，相似相溶

②溶度参数 δ 相近——热力学原则。

③表面张力相近——从胶体化学角度，形成稳定的界面层

上述3个是热力学原则

④扩散能力相近——分子动力学原则。侧链的长短相近，利于渗透和扩散，易形成浓度变化较为对称的界面扩散层。

⑤等粘性原则——流变学原则，指两相粘度接近，易混合均匀

后面两个是动力学原则

B. 共混体系组成对 η 的影响

两相共混物粘度

①理想共混体剪切粘度符合对数加和公式

Utracki公式:

$$\lg \eta_m = \omega_1 \lg \eta_1 + \omega_2 \lg \eta_2$$

- ② 正偏差——界面作用强;
- ③ 负偏差——界面作用弱;
- ④ 交替偏差——结构随组成变化 (譬如从连续相变为分散相)。

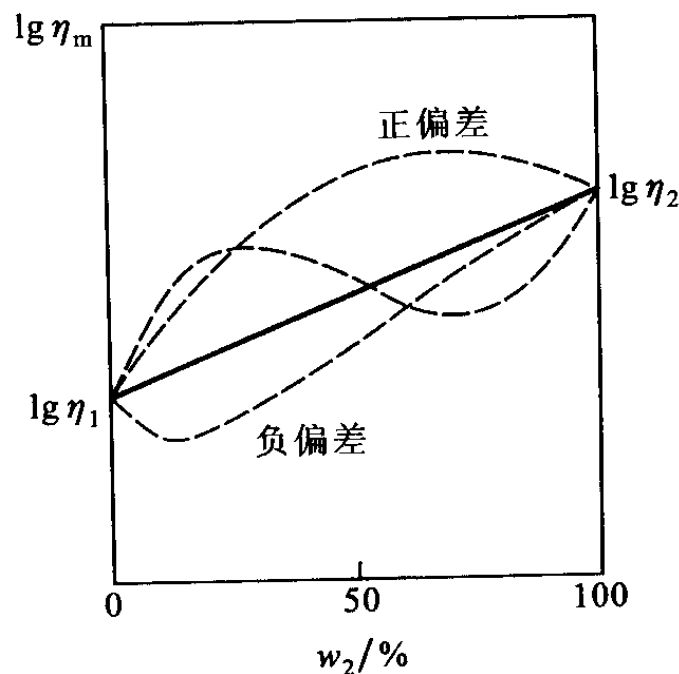


图 10-9 Utracki 简单共混物的
粘度变化规律及其偏差

案例1：不相容共混体系组成对粘弹性的影响

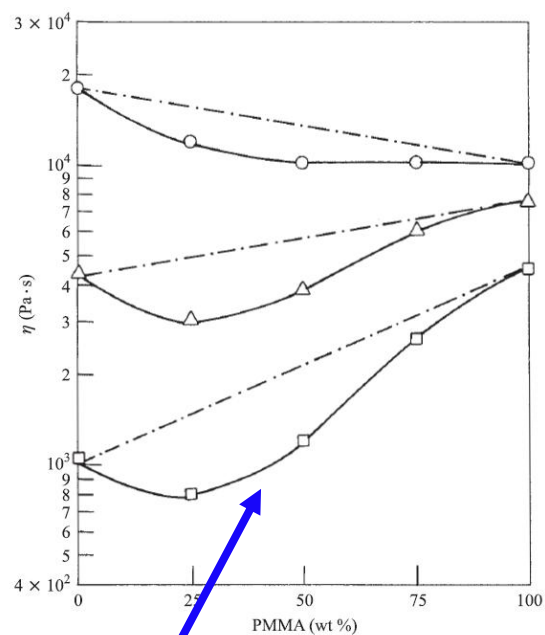


Figure 11.12 Plots of η versus blend composition for PS/PMMA blends at 200 °C at various wall shear stresses (Pa): (○) 10^5 , (△) 3×10^4 , and (□) 5×10^3 . (Reprinted from Han et al., *Applied Polymer Science Symposia* 20:191. Copyright © 1973, with permission from John Wiley & Sons.)

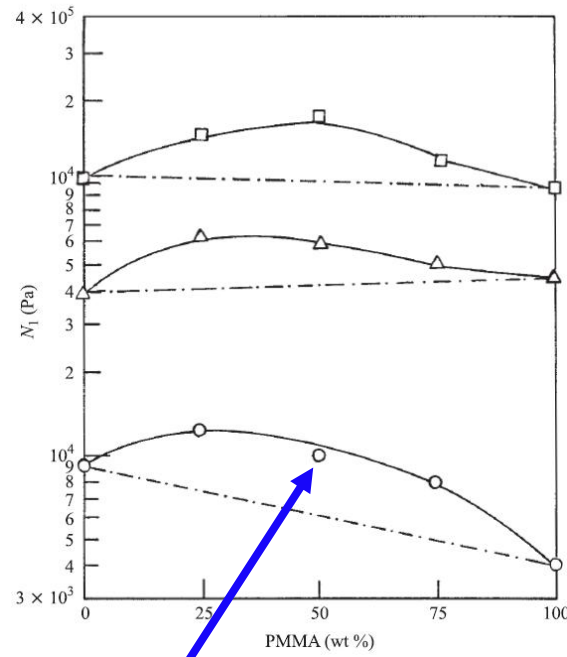


Figure 11.13 Plots of N_1 versus blend composition for PS/PMMA blends at 200 °C at various wall shear stresses (Pa): (○) 10^5 , (△) 3×10^4 , and (□) 5×10^3 . (Reprinted from Han et al., *Applied Polymer Science Symposia* 20:191. Copyright © 1973, with permission from John Wiley & Sons.)

共混物粘度（比纯聚合物）下降，但是弹性（比纯聚合物）增强

案例2：不相容共混体系组成对粘弹性的影响

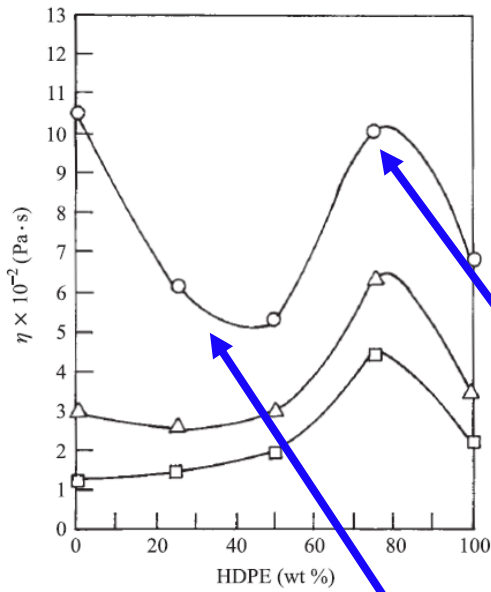


Figure 11.14 Plots of η versus blend composition for HDPE/PS blends at 200 °C at various wall shear stresses (Pa): ($\square$) 4×10^4 , ($\triangle$) 6×10^4 , ($\circ$) 9×10^4 . (Reprinted from Han and Kim, *Transactions of the Society of Rheology* 19:245. Copyright © 1975, with permission from the Society of Rheology.)

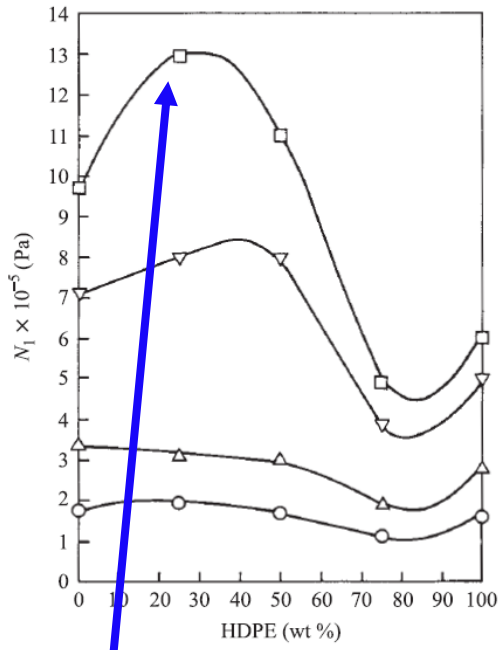
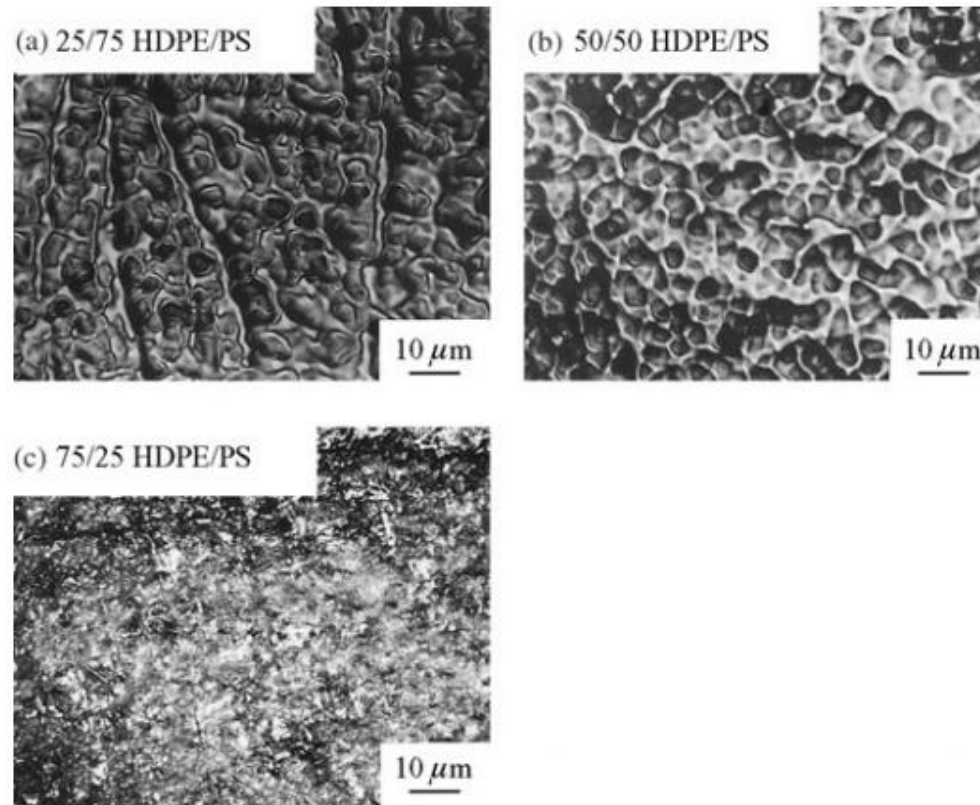


Figure 11.15 Plots of N_1 versus blend composition for HDPE/PS blends at 200 °C at various wall shear stresses (Pa): ($\circ$) 4×10^4 , ($\triangle$) 6×10^4 , (∇) 9×10^4 , and ($\square$) 10^5 . (Reprinted from Han and Kim, *Transactions of the Society of Rheology* 19:245. Copyright © 1975, with permission from the Society of Rheology.)

共混物粘度从负偏差变为正偏差，弹性与粘性的趋势正好相反

共混组成对相形态的影响



The morphology of the 75/25 HDPE/PS blend looks as if the two phases are interlocked, making the 75/25 HDPE/PS blend very difficult to flow compared with the other two blends, 25/75 and 50/50 HDPE/PS, which have a discrete HDPE phase dispersed in the continuous PS phase.

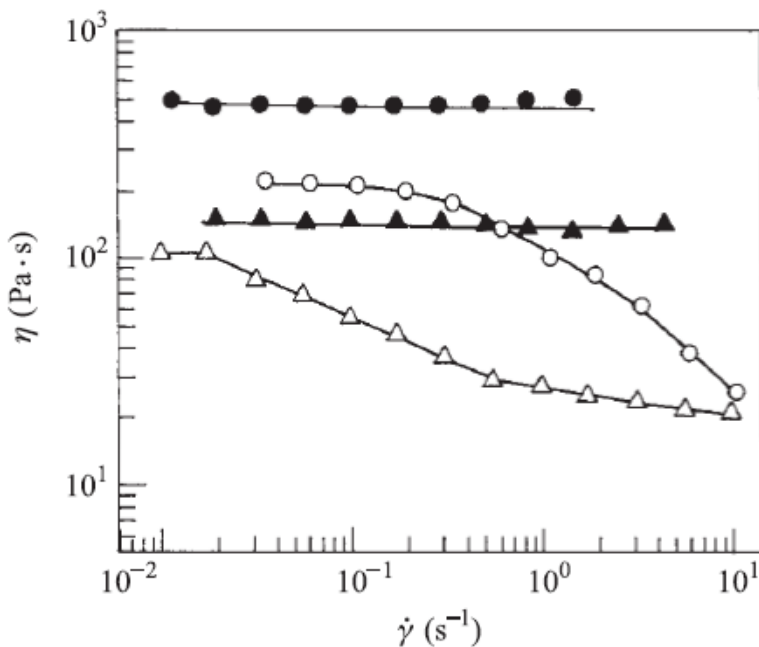
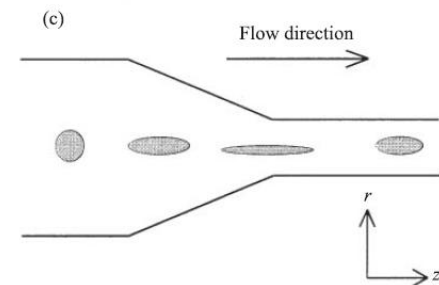


Figure 11.30 Plots of $\log \eta$ versus $\log \dot{\gamma}$ for nylon 6 (●), PET (▲), 70/30 nylon 6/PET blend (○), and 30/70 nylon 6/PET blend (△) at 270 °C. The weight-average molecular weight (M_w) and polydispersity index (M_w/M_n) determined by gel permeation chromatography are 3.1×10^4 and 2.3 for nylon 6, and 5.7×10^4 and 4.0 for PET.

- Hyman and Skalak (1972) calculated the shapes of a string of Newtonian drops, dispersed in another Newtonian liquid, moving along the centerline of a cylindrical tube and observed non-Newtonian effects arising from the deformation of drops.

两个牛顿流体共混后成为剪切变稀流体，
这是由于分散相液滴剪切变形导致垂直于
流线的截面积变小了

说明：相形态也显著影响流变行为



7.5.2. 触变剂(thixotropic agent)

(1) 触变剂的作用：防止树脂在施工的斜面或垂直面上流淌，避免树脂含量在上下层不均匀现象，从而保证制品的质量。

问题：哪些制品中要加触变剂？

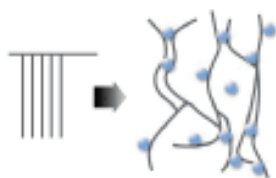
触变剂应该有什么样的结构？

常用的触变剂：气相二氧化硅、沉淀二氧化硅。其它的触变剂有石棉、高岭土、凹凸棒土、乳液法氯乙烯化合物等。

用量：0.1%-4%，根据具体要求确定

原理

补强填料



在橡胶中添加REOSIL后，部分橡胶子就会缠绕在REOSIL的表面，限制橡胶分子之间的相互作用。这样可增强橡胶强度

吸附、吸收载体



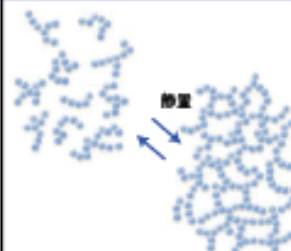
REOSIL将有效成分吸附到表面。或将其吸收到粒子之间的空隙和细孔中，这样能保持住各种有效成分

抗粘连



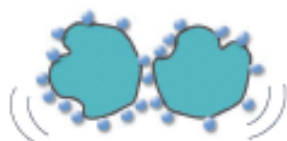
在各种树脂及高分子薄膜中加入REOSIL后，可以防止其表面相互粘结，易于剥离

增粘、触变性



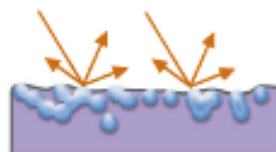
在添加了REOSIL的溶液体系里，有施加外力后流动性变大，静置后则会变硬的性质，这一性质有助于防止溶液体系的流挂

流动性



用REOSIL覆盖其它粉体颗粒的表面，可以改变其表面的性质，从而能调整流动性

消光性



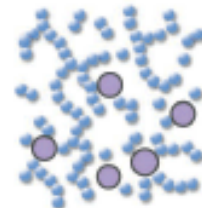
在涂料、油墨中添加REOSIL，这样在溶剂蒸发后就会在表面形成微小的凹凸面。光线在这些凹凸面上出现散射，达到抑制光亮的效果

研磨



充分分散在液体中的REOSIL粒子可对表面进行精细微研磨

防止沉淀



利用REOSIL所形成的三维网眼结构，防止溶媒中的颜料沉淀

7.5.3 增稠剂 (Thickening agent)

增稠剂是一种流变助剂，不仅可以使涂料增稠，防止施工过程中出现流挂现象，而且能赋予涂料优异的机械性能和贮存稳定性。

常用的增稠剂：无机增稠剂(膨润土)、纤维素类、聚丙烯酸酯、缩合型聚氨酯和聚多糖等。

7.5.4 填充体系的流变行为

填充体系

实际中：高分子经常都加填料，成多组分材料（降低成本、提高性能）

刚性物质①微米粒子填充②纳米粒子填充

填充体系的流变行为：

- ① 一般，随填充量增大，体系粘度增大，且常出现屈服现象。
- ② 非牛顿型流动性（剪切变稀）更为显著。
- ③ 个别高填充体系，如含填料的PA6体系，表现出剪切增稠特征。

A. 填充对流变行为的影响

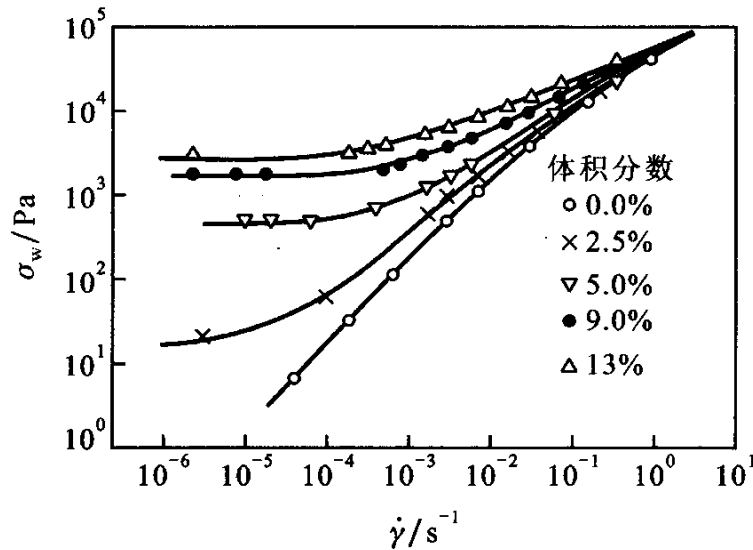


图 10-19 不同浓度的炭黑填充聚异丁烯(PIB)
在 60℃ 下剪切应力随剪切速率的变化

高填充含量的炭黑导致体系
形成网络结构，有屈服应力
如何得出这个结论的？

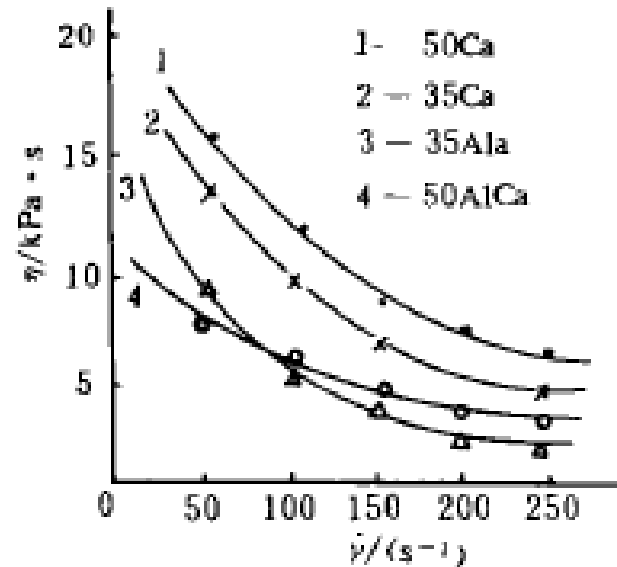


图 4 220℃ 下 ABS/CaCO₃ 体系的
粘度和剪切速率的关系

1, 2 未加偶联剂

3, 4 为加入铝酸酯偶联剂

偶联剂加入促进填料分散，
有利于在相同粘度下提高填料用量

B. 填充体系的粘度公式

1) 低填充量时，填充体系粘度可用Einstein公式：

$$\eta_r = \eta/\eta_0 = 1 + K_E\phi$$

式中： η_0 为牛顿型分散介质的粘度， K_E 为Einstein系数：
对刚性球形粒子， $K_E = 2.5$ ；若为刚性棒状粒子，且长径比 >1 ， $K_E > 2.5$ 。 ϕ 为刚性分散粒子的体积分数

特性粘数和流体力学半径

$$[\eta] = \lim_{c \rightarrow 0} \frac{\frac{\eta}{\eta_s} - 1}{c}$$

η_s : 溶剂的粘度

η : 溶液的粘度

c : 溶液中聚合物的质量浓度

作业题1: 根据Einstein方程, 推导特性粘数和分子量的关系

B. 填充体系的粘度公式

2) 当填充量增加，需要考虑填料间相互作用，Guth-Gold方程提供更准确描述：

$$\eta_r = 1 + 2.5\phi + 14.1\phi^2$$

对于非球形填料，需引入形状因子f

$$\eta_r = 1 + 2.5\phi + 14.1\phi^2 f^2$$

3) 刚性粒子浓度较大时（高填充量），Mooney公式：

$$\ln(\eta/\eta_0) = K_E \phi / (1 - \phi/\phi_m)$$

M. Mooney, *J. Colloid Sci.*,
6, 162–170 (1951).

ϕ_m 为分散相最大堆砌系数

粒子有序堆砌聚集时， $\phi_m = 0.74$ ；

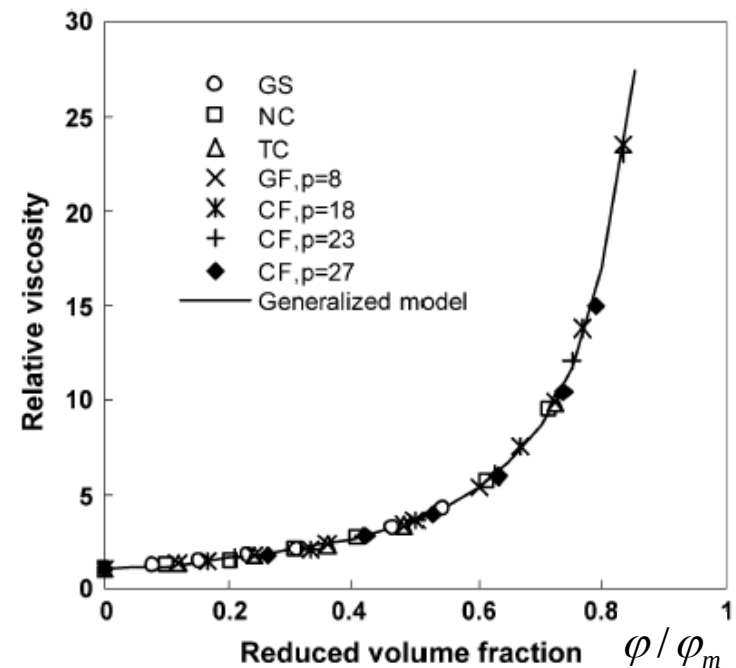
粒子随机分布未聚集时， $\phi_m = 0.60$ - 0.63 ；

粒子随机聚集时， $\phi_m = 0.37$ 。

Parameters of relative viscosity for various fillers from Kiatno's results

Filler	Adjustable parameter B in Eq. (3) (extracted from [24])	ϕ_m (estimated values using graphical method)
Glassy sphere (GS)	0.68	0.645
Natural calcium carbonate (NC)	0.52	0.49
Talc (TC)	0.44	0.415
Glass fiber (GF with $p = 8$) ^a	0.44	0.415
Carbon fiber (CF with $p = 18$)	0.315	0.3
Carbon fiber (CF with $p = 23$)	0.253	0.24
Carbon fiber (CF with $p = 27$)	0.2	0.19

^a p is the aspect ratio of fiber.



R. Arefinia, A. Shojaei / *Journal of Colloid and Interface Science* 299 (2006) 962–971

C. 纤维长径比对纤维增强聚合物复合材料性能的影响

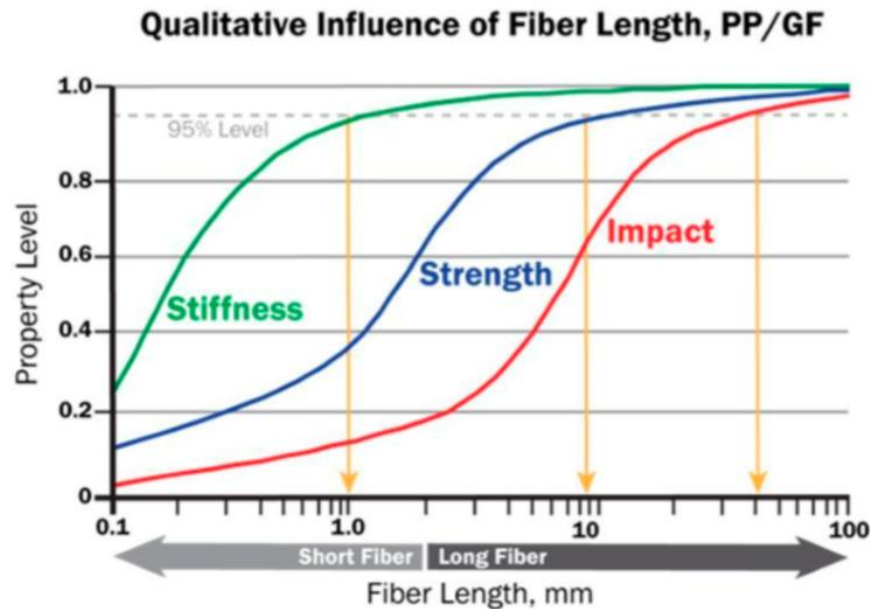


Figure 1. Dependence of properties with fiber length in glass fiber reinforced polypropylene (PP)
Reprinted with permission from [15].

纤维长径比增加有助于提高力学性能，
但是纤维无序取向体系的粘度也会显著提高。

作业题2：如何克服高纤维填充量导致粘度增加而难加工的问题？

案例：聚丙烯填充碳酸钙

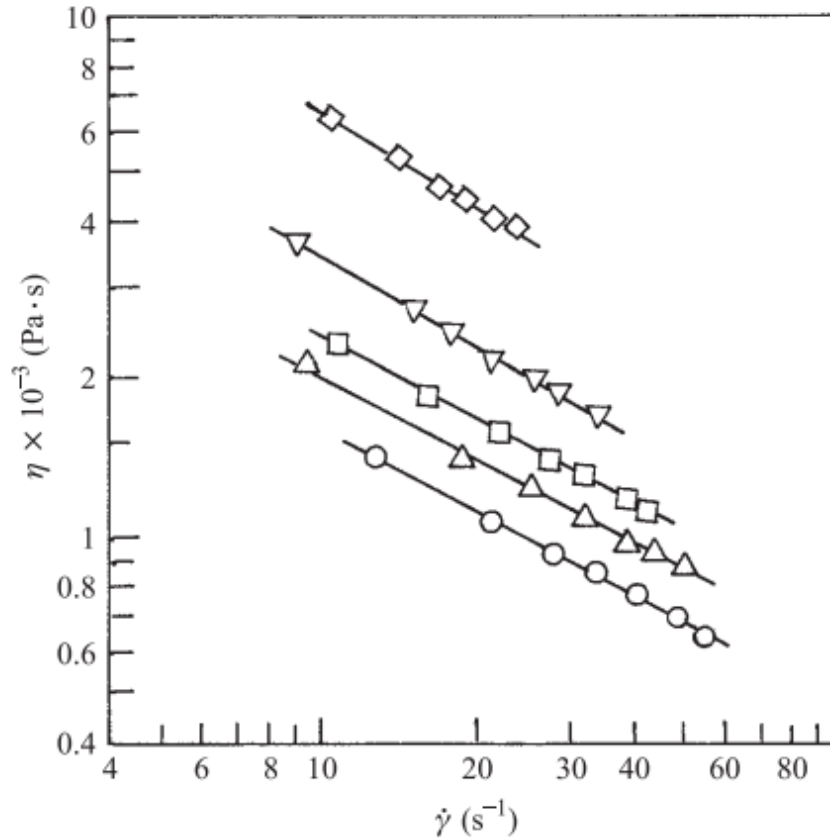


Figure 12.2 Plots of $\log \eta$ versus $\log \dot{\gamma}$ at 200 °C for neat PP (○) and PP composites filled with varying amounts of CaCO_3 (wt %): (△) 10, (□) 20, (▽) 40, and (◇) 70. The data were obtained using a slit die with the dimensions given in Figure 12.1. (Reprinted from Han, *Journal of Applied Polymer Science* 18:821. Copyright © 1974, with permission from John Wiley & Sons.)

填充体系粘度增加

案例：聚丙烯填充碳酸钙

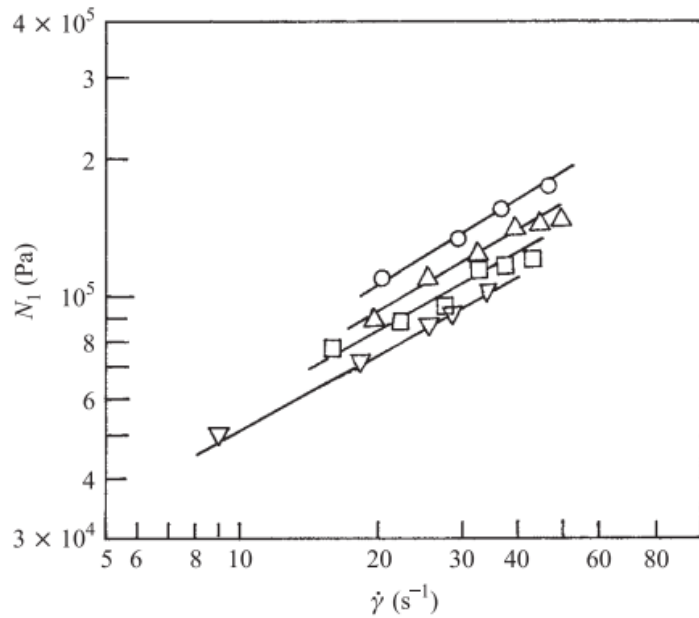
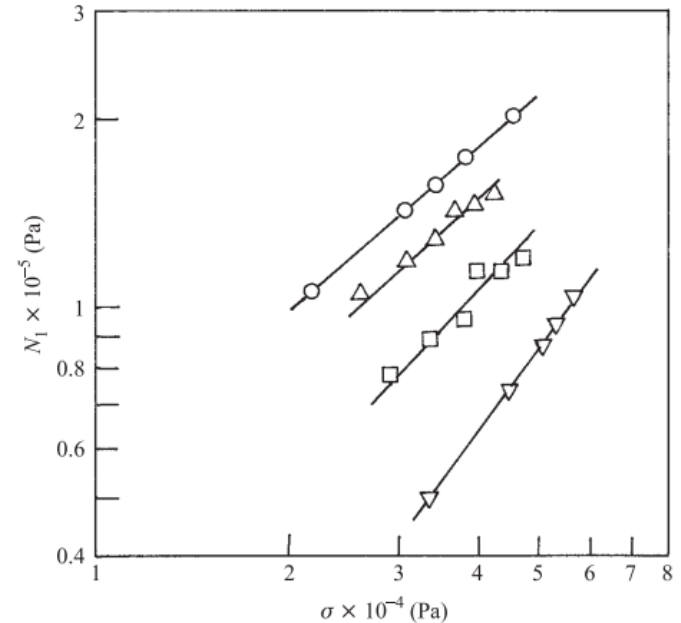


Figure 12.4 Plots of $\log N_1$ versus $\log \dot{\gamma}$ at 200 °C for neat PP (○) and CaCO₃-filled PP composites with varying amounts of CaCO₃ (wt %): (△) 10, (□) 20, and (▽) 40. The exit pressure method described in Chapter 5 was used to determine values of N_1 from a slit die with the dimensions given in Figure 12.1. (Reprinted from Han, *Journal of Applied Polymer Science* 18:821. Copyright © 1974, with permission from John Wiley & Sons.)



填充体系弹性减弱

案例：聚苯乙烯填充炭黑

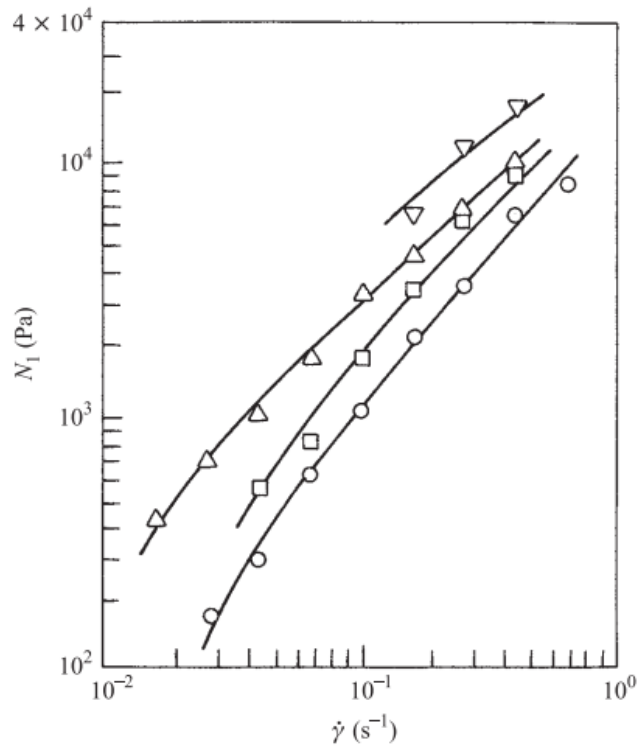
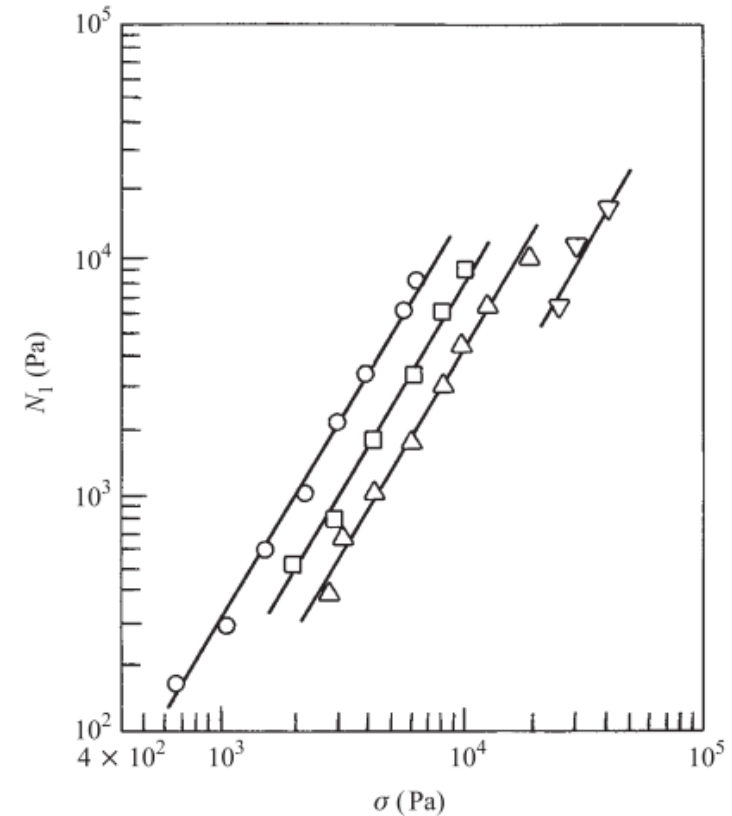


Figure 12.8 Plots of $\log N_1$ versus $\log \dot{\gamma}$ at 180 °C for neat PS (○) and carbon-black-filled PS composites with varying amounts of carbon black (wt %): (□) 10, (△) 20, and (▽) 30. The data were obtained using a cone-and-plate rheometer. (Reprinted from Tanaka and White, *Polymer Engineering and Science* 20:949. Copyright © 1980, with permission from the Society of Plastics Engineers.)



该填充体系弹性是增强了还是减弱了？

D. 通过填料多分散性降低粘度

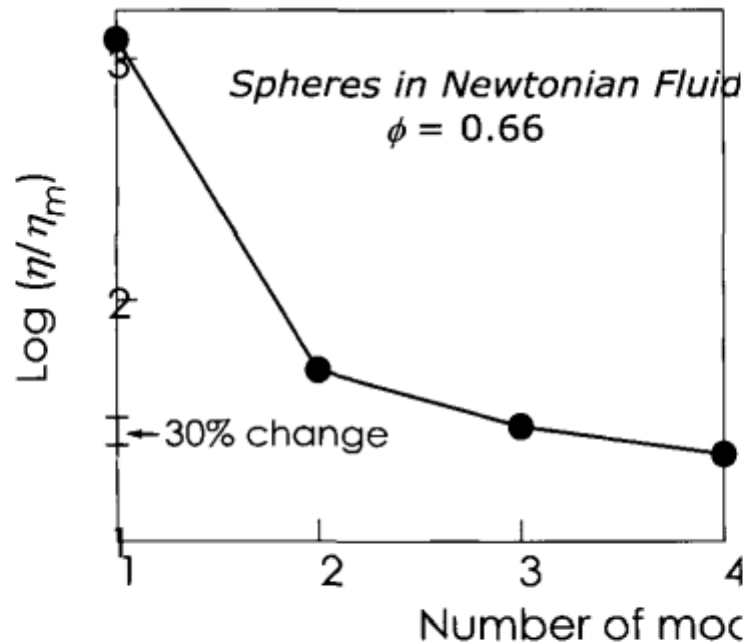


Figure 12-1. Illustration of the optimal reduction in viscosity by using a bimodal, trimodal and tetramodal distributions of spherical particles at a fixed volume fraction of 0.66. The sizes of the modes are optimized to fit into the spaces between the larger particles. The reduction is most dramatic on going from a unimodal to a bimodal distribution. (Data from Metzner.²)

作业

题3：对添加**相同重量分数球形填料**的聚合物熔体，微米级填料和纳米级填料，何者的熔体粘度更小？

题4：对添加相同体积分数填料的聚合物熔体，球形、棒状和片层状填料，何者的熔体粘度更小？

第8节 剪切和拉伸对聚合物结构的影响

8.1 剪切和拉伸过程中材料结构的检测方法

应力双折射（或光弹法）、粒子示踪

8.2 剪切和拉伸流动对聚合物多相体系相态结构的影响

流动导致不均匀分布

（低粘度组分趋壁、颗粒趋于管中间）

组分含量和粘度比影响何者是分散相；

毛细管数决定分散相粒径

8.3 剪切取向

8.4 拉伸取向

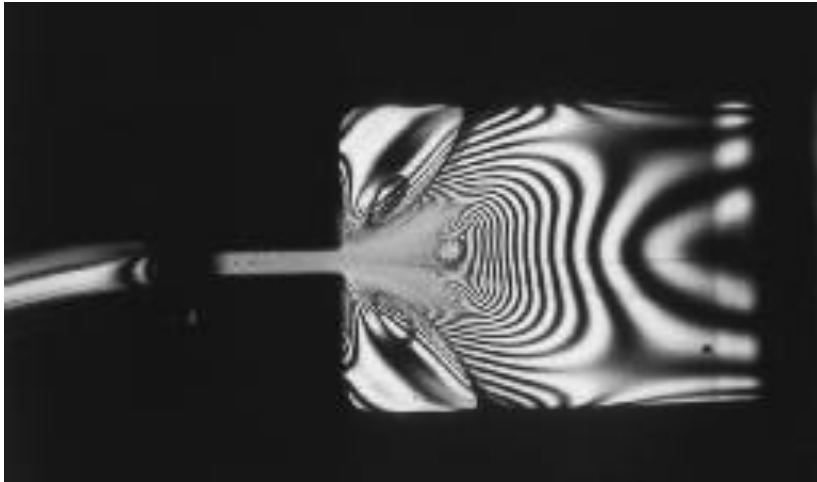
8.5 剪切流动对结晶的影响

（剪切速率和剪切应变超过某一临界值，促进结晶）

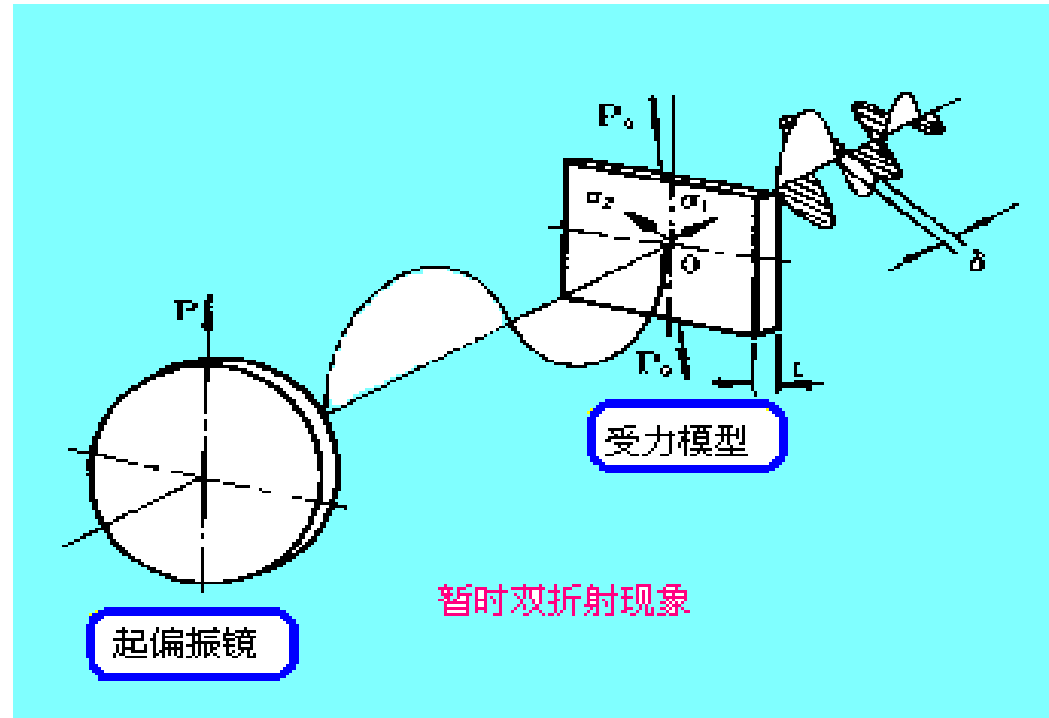
（结晶是否取向，取决于结晶速率与熔体松弛速率之比）

8.1 流动过程中材料结构的检测方法

1. 应力双折射法（可以测定制件中的残余应力或者流动熔体中的主应力差）



无定型聚合物产生双折射的原因：
残余应力或取向



应力双折射定律:

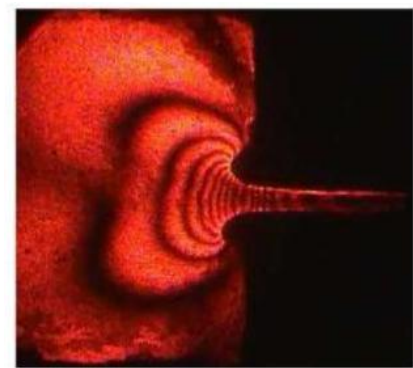
$$\delta = ct(\sigma_1 - \sigma_2)$$

δ —— 光程差;

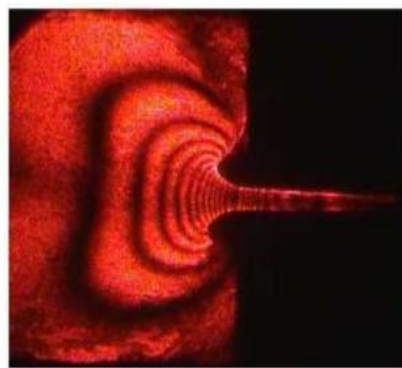
c —— 应力光学系数, 是材料的常数

t —— 样品厚度。

$(\sigma_1 - \sigma_2)$ —— 某点的主应力差。



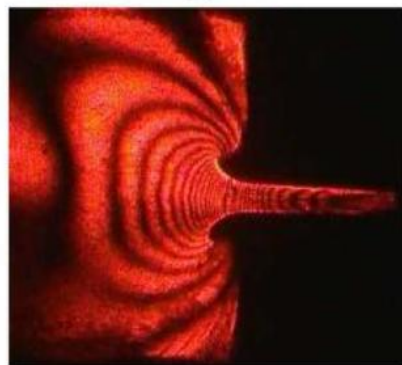
(a)



(b)



(c)

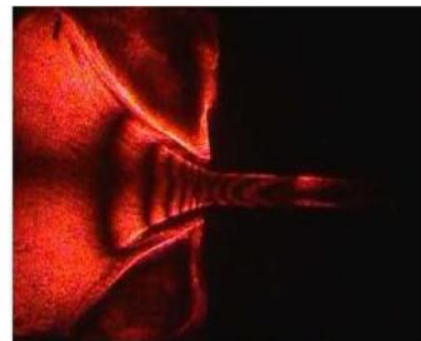


(d)

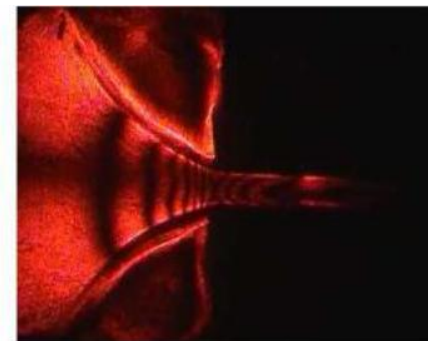
图 4-3 HDPE TR144 在不同螺杆转速时的双折射图

(a) 15 rpm; (b) 20 rpm; (c) 25 rpm; (d) 30 rpm

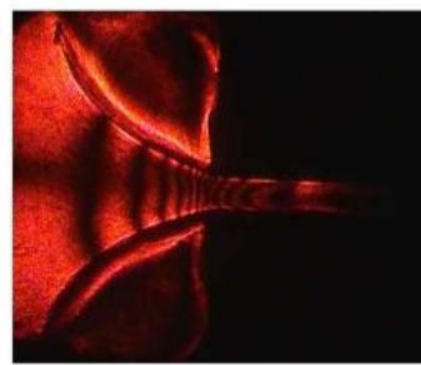
Fig. 4-3 Birefringence fringes at different screw rotation speed of HDPE TR144



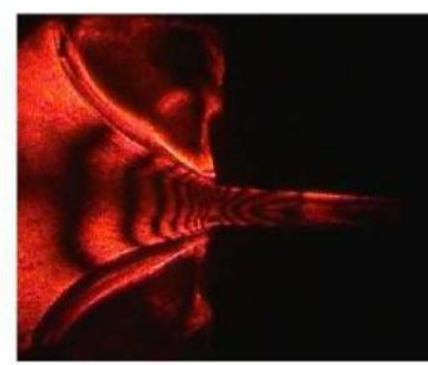
(a)



(b)



(c)



(d)

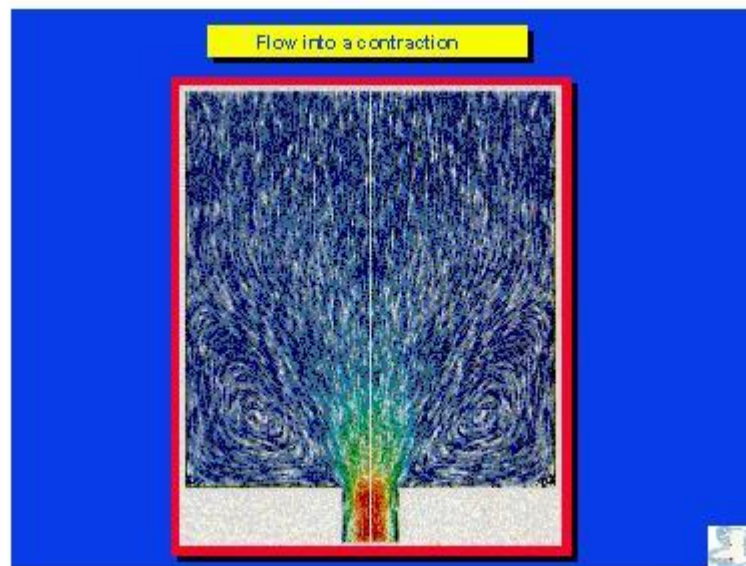
图 4-4 LDPE 2426K 在不同螺杆转速时的入口长度双折射图

(a) 15 rpm; (b) 20 rpm; (c) 25 rpm; (d) 30 rpm

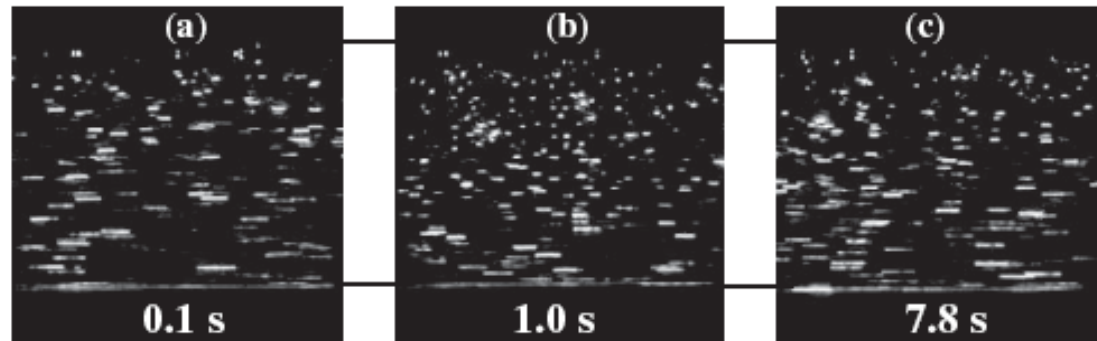
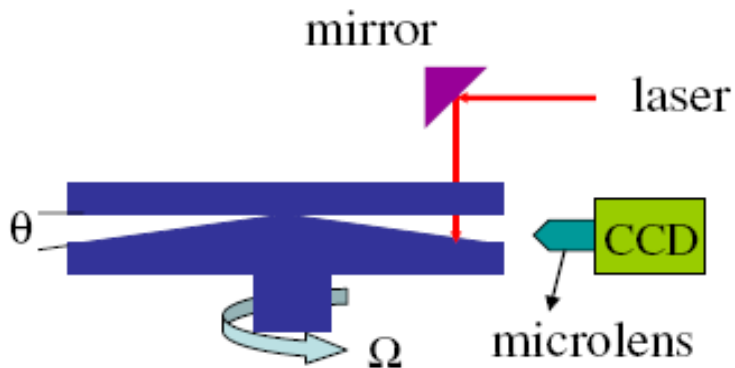
Fig. 4-4 Birefringence fringes from the entrance at different screw rotation speed of LDPE

来源：华南理工大学范吉昌博士学位论文《聚乙烯熔体在突变收敛流道挤出口模内流变行为双折射理论与实验研究》，指导教师：彭响方，2011.

2. 粒子示踪法



如何保证测量结果的正确性？



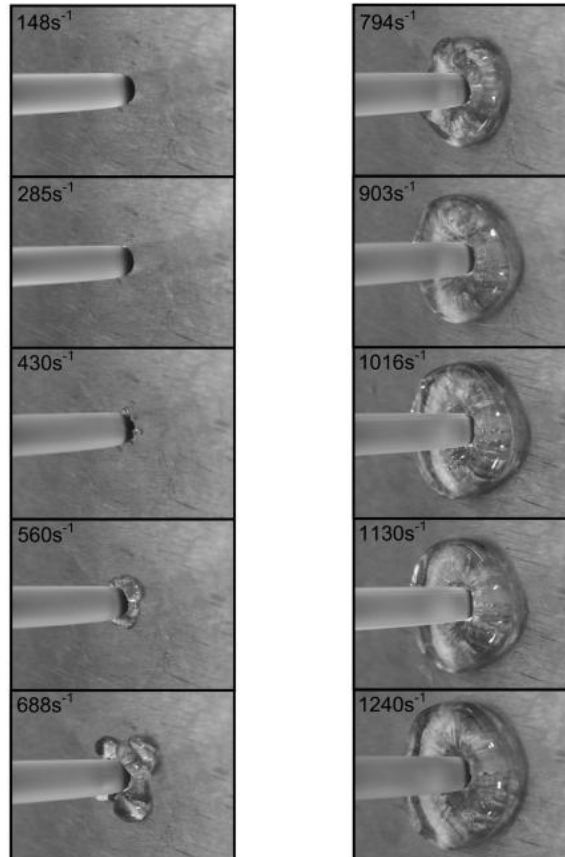
The entangled polymer solution under study is similar to a previously prepared and studied 10 wt% 1,4-polybutadiene (PBD) solution [30], made of a high weight PBD of $M_w = 1.24 \times 10^6$ g/mol and $M_w/M_n = 1.18$ from Polymer Source Inc., dissolved in a phenyl-terminated oligomeric butadiene of $M_n = 1.0$ K (Aldrich

silver-coated particles of $10 \mu\text{m}$ diameter (Dantec Dynamics HGS-10) at a low level of several hundred parts per million. A dynamic mechanical spectrometer (Advanced Rheometrics Expansion System–ARES) equipped with a cone-plate assembly of 25 mm diameter was employed to make velocity profile measurements, where the cone angle θ is 5° .

Prashant Tapadia and Shi-Qing Wang. PHYSICAL REVIEW LETTERS 96, 016001 (2006)

8.2 剪切和拉伸流动对聚合物多相体系相态结构的影响

8.2.1 挤出过程中低分子量组分趋于管壁分布



Die exit visualization after 10 min of extrusion of HDPE Liten FB 29 2961 at different apparent shear rates at 170 °C.

8.2.3 剪切改变相分离温度

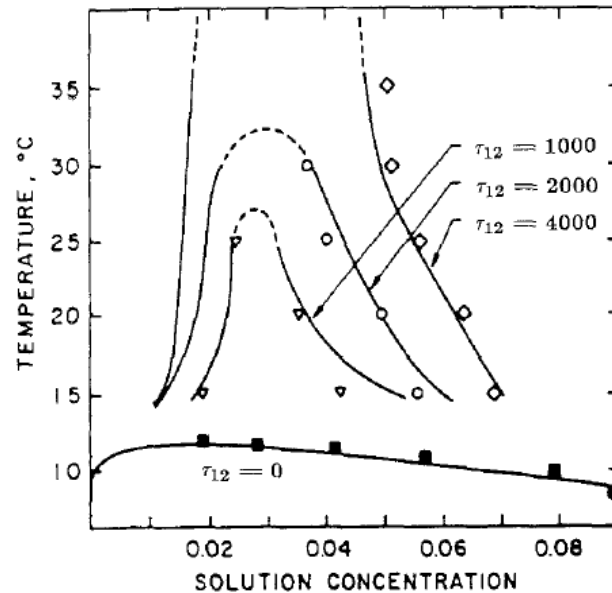


Fig. 4. "Stability diagram" for polystyrene in dioctylphthalate. The points give the experimental cloud points and the lines are the theoretical predictions. The shear stress τ_{12} is given in dyn/cm², and the concentration is given in g/mL. (From Rangel-Nafaile, 1984)

8.2.3 剪切和拉伸流动对聚合物液/液混合体系结构的影响

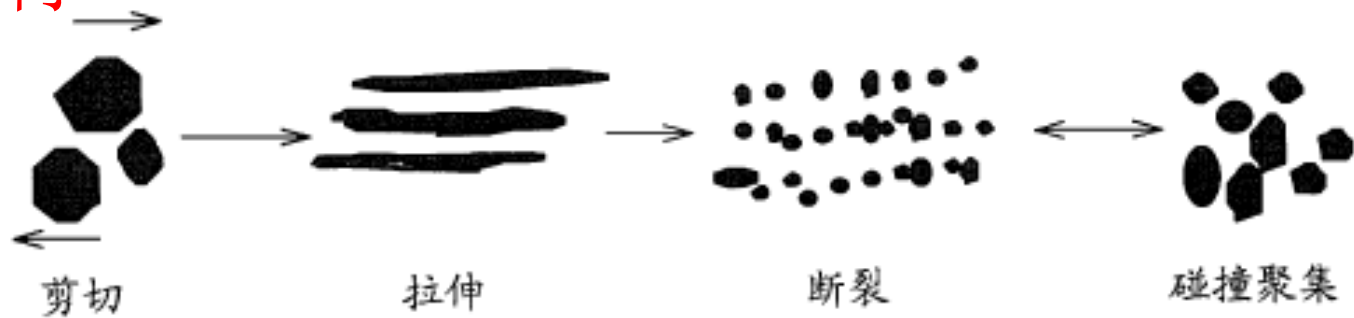


图1.2 Elemans提出的聚合物共混结构发展机理示意图

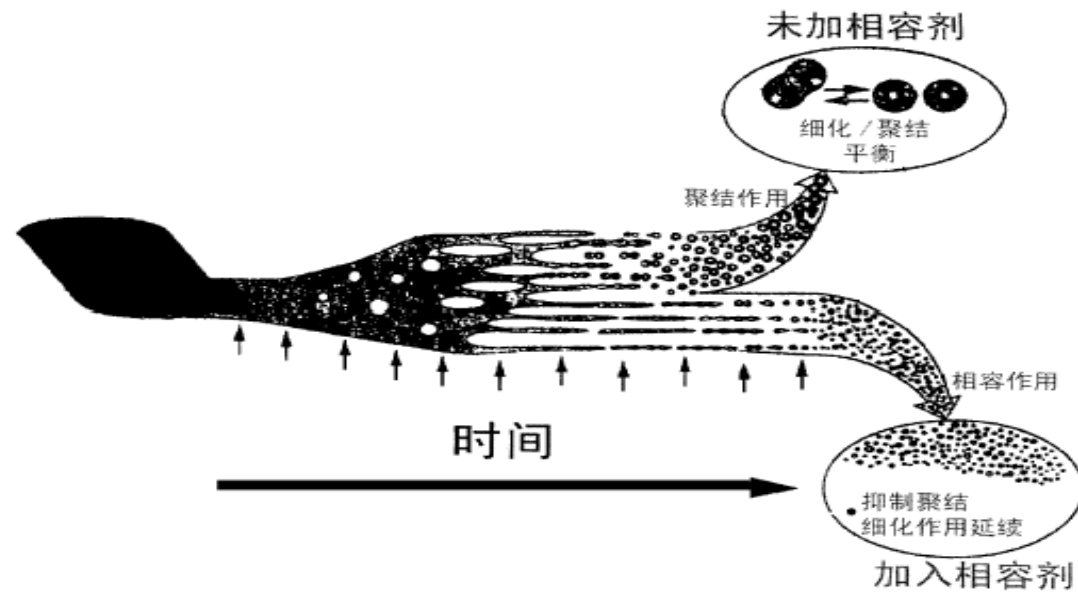
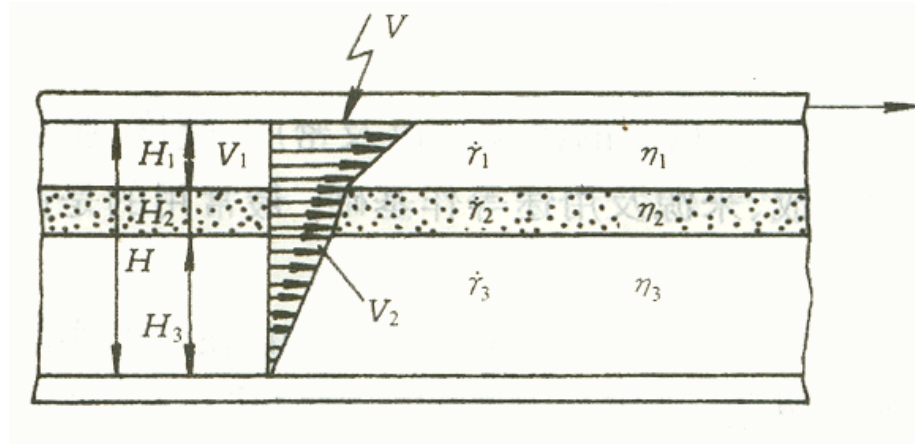


图1.3 Macosko提出的聚合物共混初期结构发展机理示意图

A. 流变性非均匀流体的层流混合



图：流变性不均匀流体在平行平板混和器中的混和

不同流体内部剪切应力相等，低粘度流体内部剪切速率大，相同时间内剪切应变更大，容易分散；高粘度流体内部剪切速率小，相同时间内剪切应变更小，不容易分散

高粘度的少组分混合到低粘度的多组分中-----比较困难
低粘度的少组分混合到高粘度的多组分中-----相对容易

B. 粘度比和界面张力对液体分散的影响

Mode of liquid droplet deformation in uniform shear field

	Mode of Deformation	Range of Viscosity Ratio and Surface Tension
Uniform Shear Flow		$k < 0.14, \quad \sigma < 10$
		$0.14 < k < 0.65, \quad 10 < \sigma < 20$
		$0.7 < k < 2.2, \quad \sigma < 20$
		$k > 3.8, \quad \sigma > 4.0$

F. D. Rumscheidt and S. G. Mason, *J. Colloid Sci.* **16**, 238 (1961).

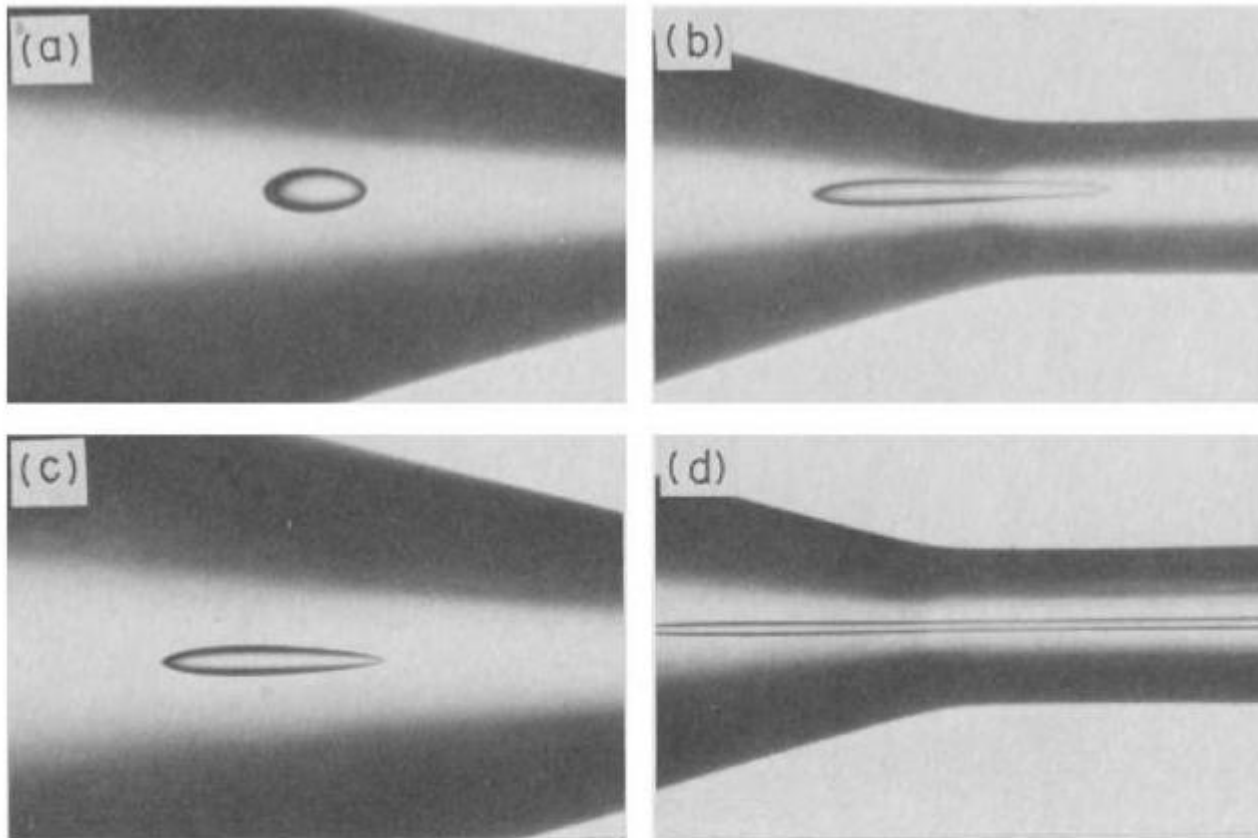


Fig. 4.14 Photographs of droplets describing the effect of shear rate on the extent of droplet deformation [52]: (a) and (b) at $\dot{\gamma} = 13.7 \text{ sec}^{-1}$; (c) and (d) at $\dot{\gamma} = 128.7 \text{ sec}^{-1}$. The droplet phase is a 6% PIB solution, the suspending medium is a 2% Separan solution, and the initial droplet size is 1.20 mm. From H. B. Chin and C. D. Han, *J. Rheol.* **23**, 557. Copyright © 1979. Reprinted by permission of John Wiley & Sons, Inc.

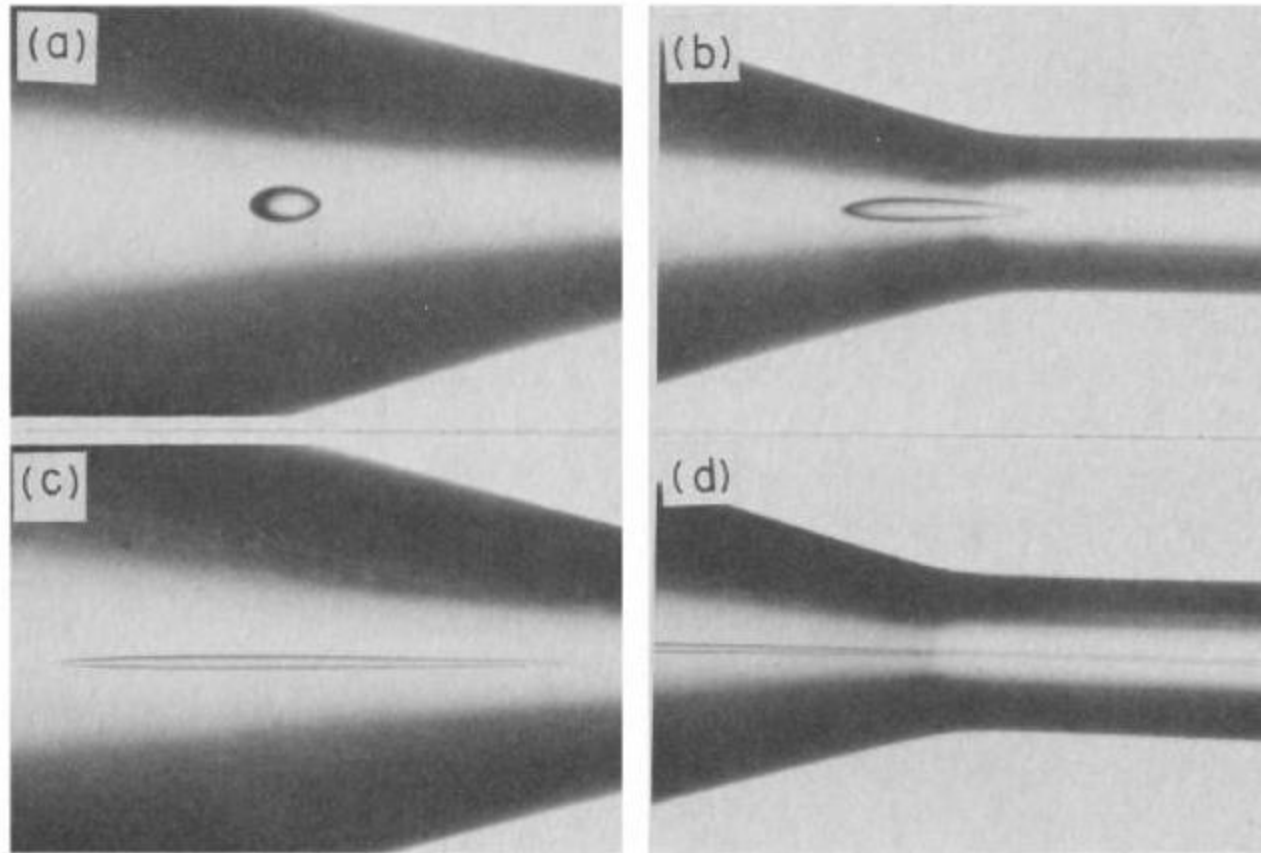


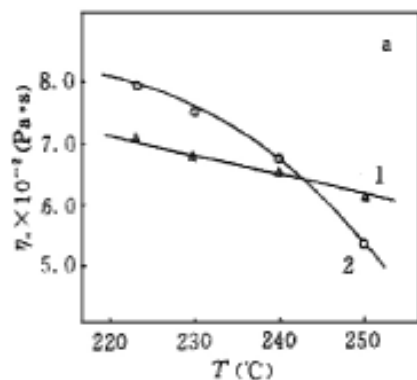
Fig. 4.15 Photographs of droplets describing the effect of suspending medium viscosity on the extent of droplet deformation [52]: (a) and (b) with a 2% Separan solution; (c) and (d) with a 4% Separan solution. The droplet phase is a 6% PIB solution, the initial droplet size is 0.91 mm, and the apparent wall shear rate is 15.0 sec^{-1} . From H. B. Chin and C. D. Han, *J. Rheol.* **23**, 557. Copyright © 1979. Reprinted by permission of John Wiley & Sons, Inc.

C. 影响液体-液体分散的主要因素

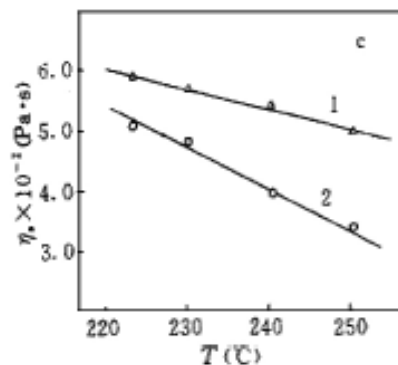
分散相液滴破裂的相关因素：

- 表面张力：表面张力越小，越易破裂；(类比弹性模量，相同应力的情况下，模量越小，应变越大)
- 分散相的黏度：分散相的黏度越大，液滴越难以破裂。研究表明，当黏度比大于4，液滴不破裂。
- 剪切应力：剪切应力越大，液滴越易破裂。低于临界剪切应力，液滴不破裂。

案例：HDPE/mPA层状阻隔合金的制备

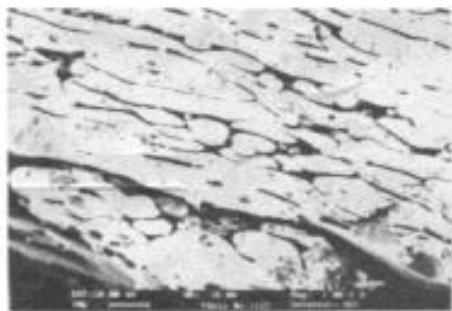


(a) $\dot{\gamma} = 200 \text{ s}^{-1}$; 1: HDPE; 2: mPA

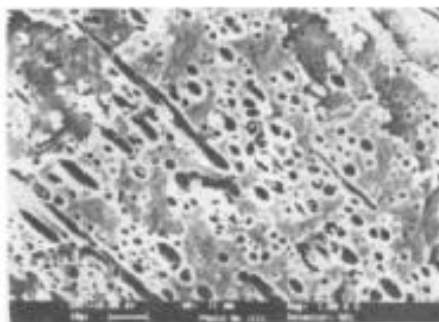


(c) $\dot{\gamma} = 400 \text{ s}^{-1}$; 1: HDPE; 2: mPA

思考：欲获得PA在HDPE基体中以层状分散的结构，应该在什么温度和剪切速率范围内加工？



(c) article at low shear rate (formic acid etch)



(e) article at high shear rate (formic acid etch)

Tab 1 Permeability rate in 230 mL blowmolded bottles for duration is 180 days at room temperature

Solvent	HDPE bottle (%)	HDPE/mPA Laminar bottle (%)	Permeability factor
ethylene	37.1	0.45	82
methylben-	43.3	0.68	63
zene	45.1	0.73	62
benzene	13.9	0.15	93
gasoline			

D. 影响液体分散相尺寸的无量纲数：毛细管数

- 共混物组分
- 流场类型及其强度
- **毛细管数** Ca ，表示作用在分散相颗粒上的剪切应力和界面张力的比值（对于分散非常重要的概念）。

$$Ca = \frac{\tau}{\frac{\sigma}{R}} = \frac{\eta_m \dot{\gamma} R}{\sigma}$$

τ : 剪切应力

σ : 界面张力

R : 分散相液滴的半径

η_m : 连续相粘度

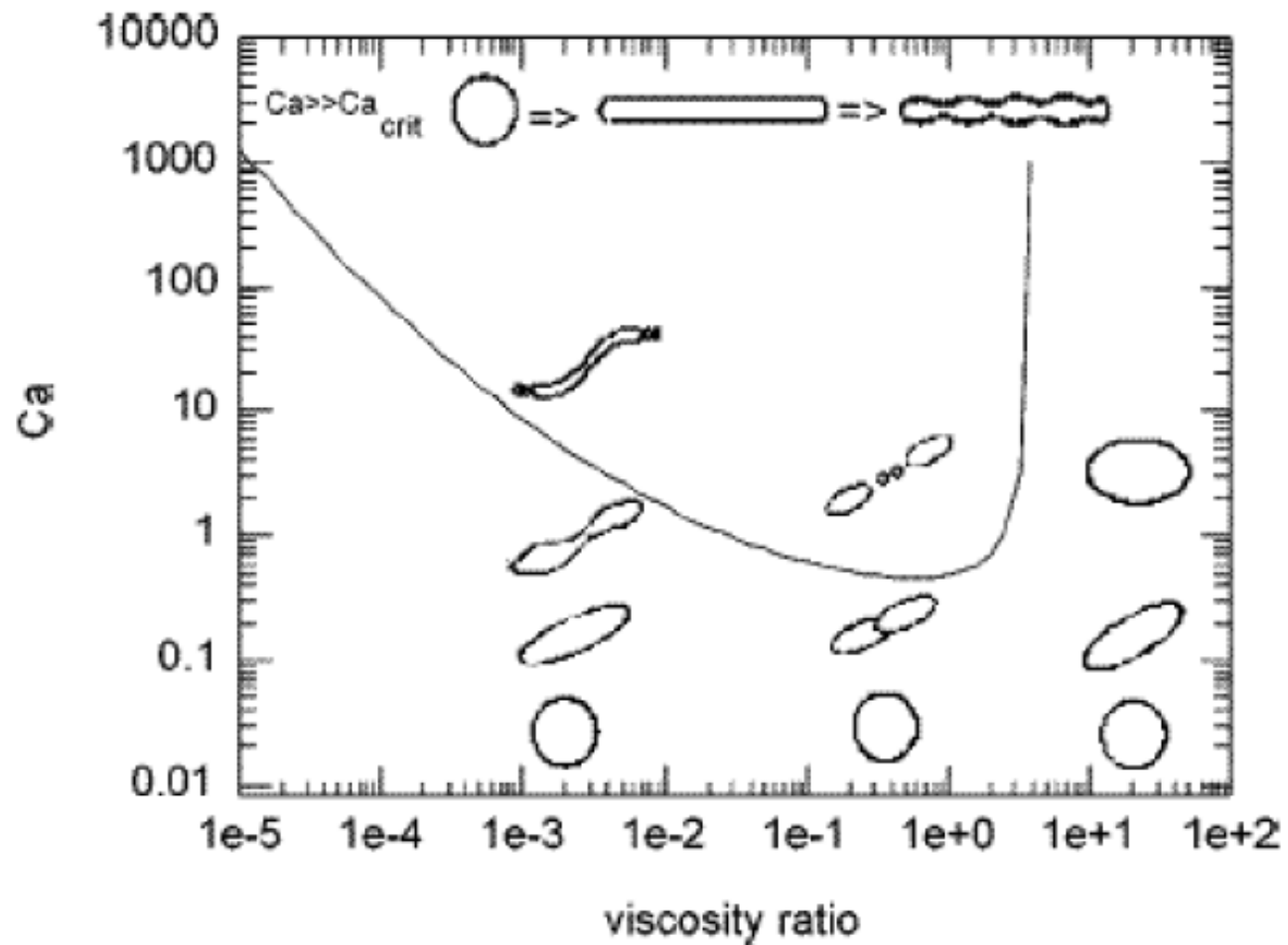


Figure 2: Schematic representation of the different modes of deformation and break-up in shear flow. The full line corresponds to an empirical fit for the critical capillary number through the data of Grace [18] (figure taken from [20]).

E. 分散相粒径的预测与计算

(1) Taylor 率先研究了简单剪切场下牛顿体系中的液滴破碎理论，给出了由临界毛细管数预测共混体系分散相最小直径的关系式：

$$D = \frac{2\sigma \cdot (Ca)_{crit}}{\eta_m \dot{\gamma}} \quad (Ca)_{crit} = \frac{1}{2} \frac{16\eta_r + 16}{19\eta_r + 16}$$

毛细管数的定义式

理论推导的临界毛细管数公式

其中：

D 为分散相颗粒的最小直径； σ 为界面张力；

$(Ca)_{crit}$ 为临界毛细管数；

η_r 为分散相和基体（连续相）的粘度之比；

η_m 为基体（连续相）的粘度

E. 分散相粒径的预测与计算

(2) Wu在Taylor推导的颗粒尺寸计算公式基础上，分别使用尼龙和聚对苯二甲酸乙二醇酯为基体，通过获得的主曲线方程得到反映共混体系界面张力和黏度比对分散相粒径影响的经验关系式：

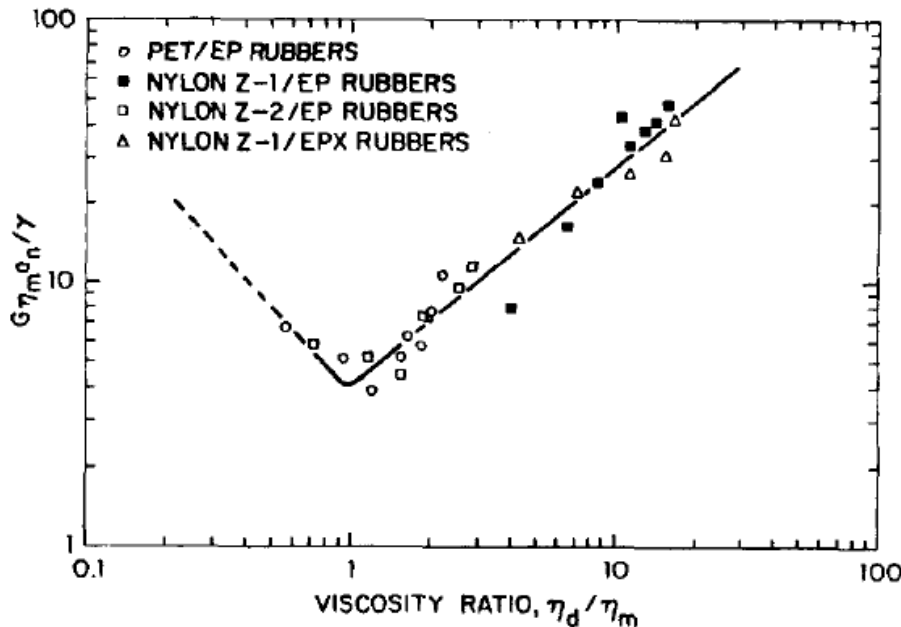


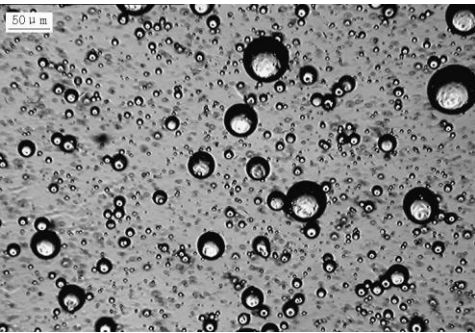
Fig. 11. Dimensionless master curve of $G\eta_m a_n / \gamma$ vs. η_d / η_m for the melt extrusion using a co-rotating twin screw extruder.

$$D_{Wu} = 4\eta_r^{\pm 0.84} \frac{\sigma}{\eta_m \dot{\gamma}}$$

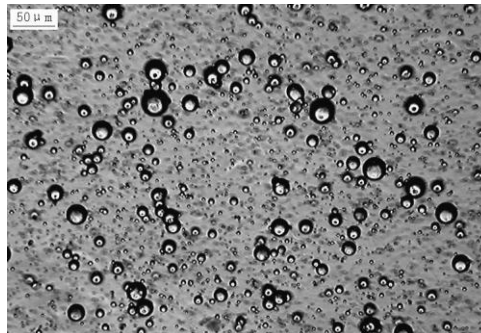
$\eta_r > 1$ 时，取正； $\eta_r < 1$ 时，取负

何时分散相粒径取最小值？

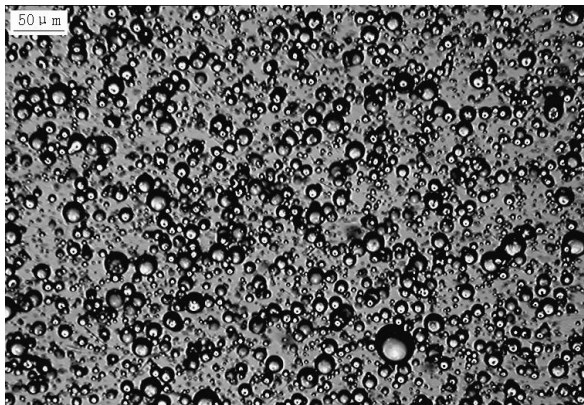
案例：剪切速率对分散相尼龙尺寸的影响



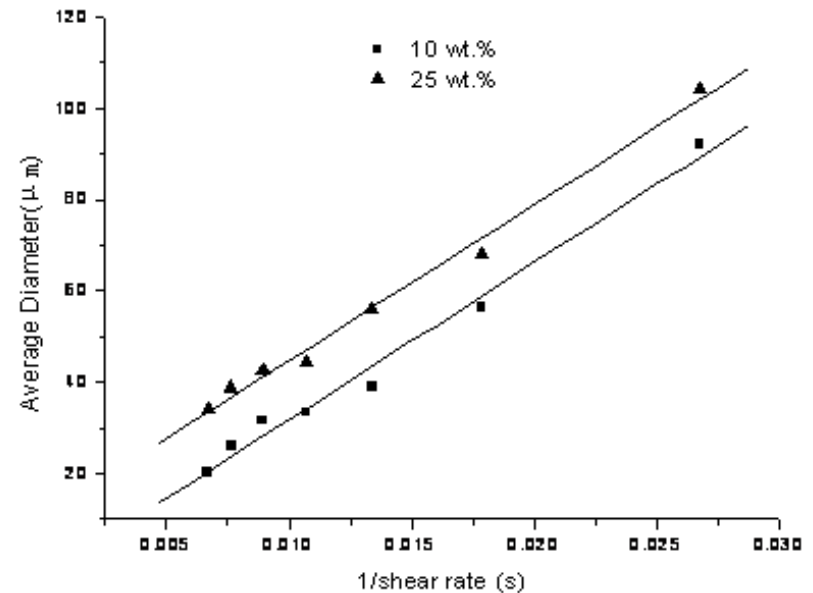
平均 $\dot{\gamma} = 37.4 \text{ s}^{-1}$



平均 $\dot{\gamma} = 93.4 \text{ s}^{-1}$



平均 $\dot{\gamma} = 149 \text{ s}^{-1}$



剪切后停止后必须及时冻结相结构，否则颗粒会聚并长大！

F. 粘度比影响共挤出的两相分布

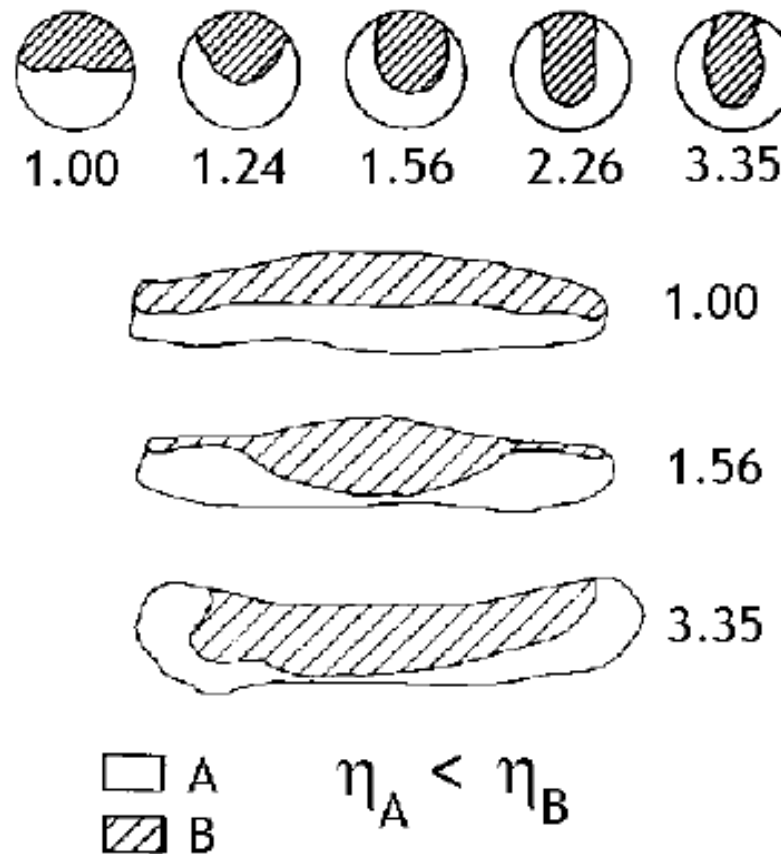


Figure 1.13. Layer-to-layer flow rearrangement as a function of time.

粘度小的组分倾向于包裹粘度大的组分

G. 根据粘度比设计共挤出的两相截面

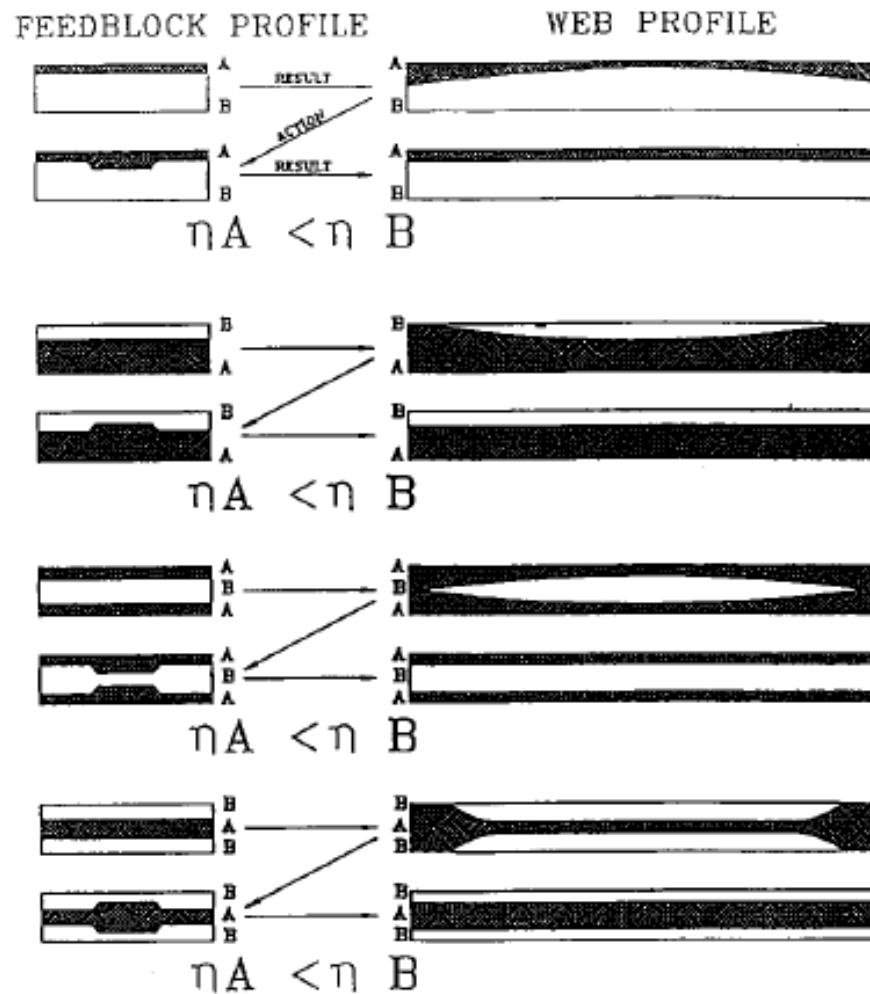
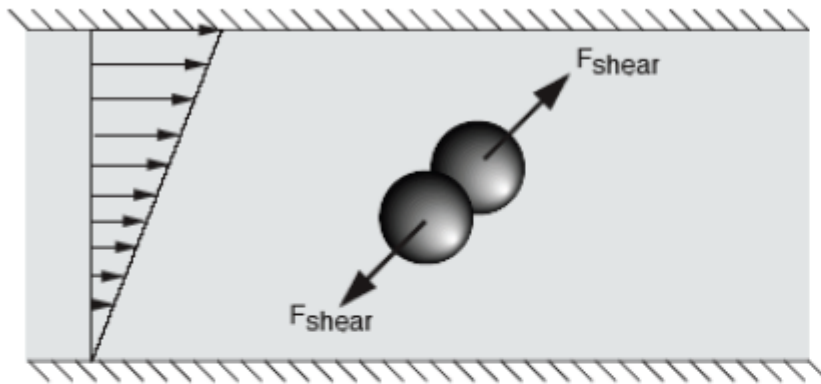


Figure 1.14. Feedblock profiling and the resultant effects.

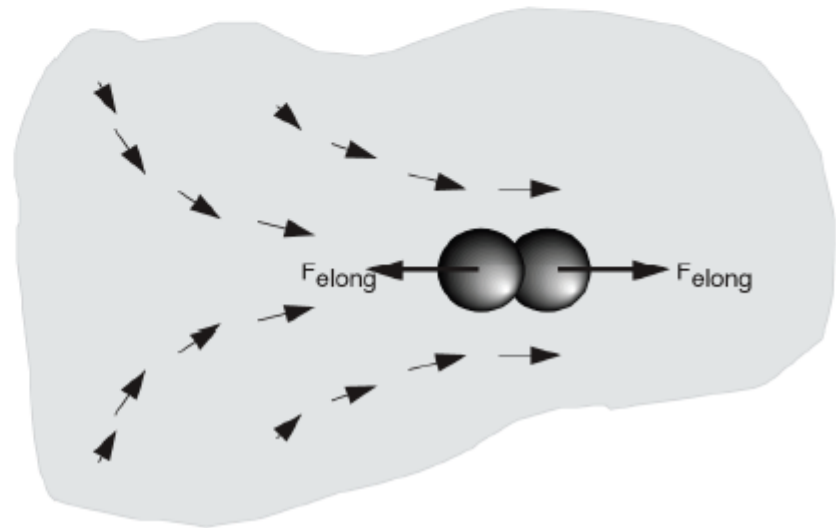
H. 拉伸的分散效果比剪切更有效！

Dispersive Mixing



$$F_{shear} = 3\pi\eta\dot{\gamma}r^2$$

$$F_{elongation} = 6\pi\eta\dot{\gamma}r^2$$



拉伸比剪切更有利于分散

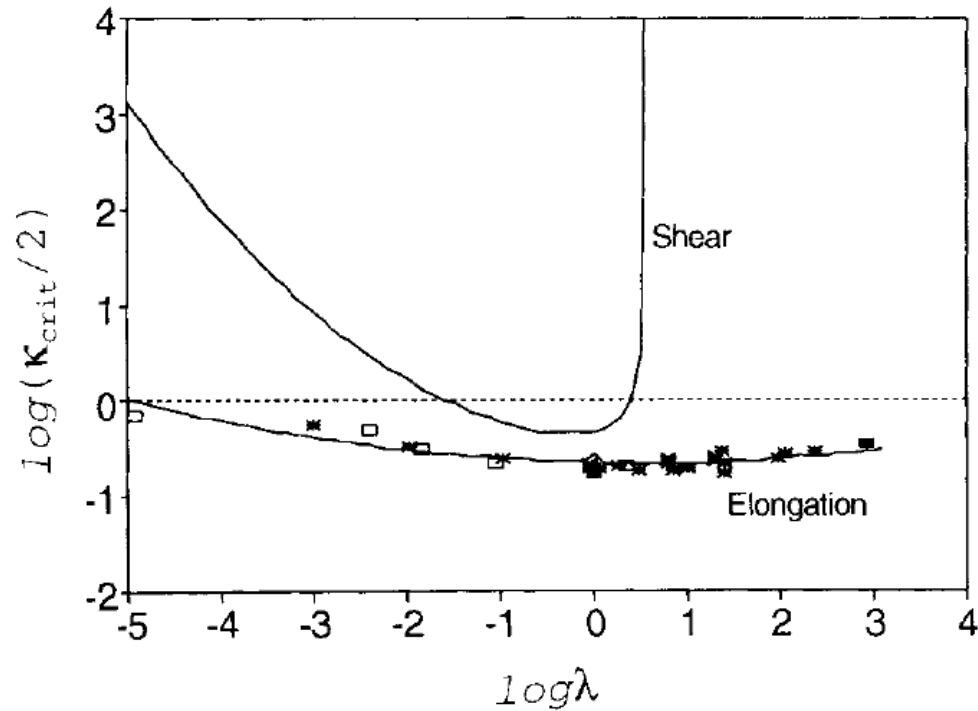


Fig. 5. Comparison of effect of viscosity ratio on critical capillary number in shear and elongational flows.

I. 小结：流动对分散相尺寸的影响

On the basis of the reviewed literature the following conclusions can be made:

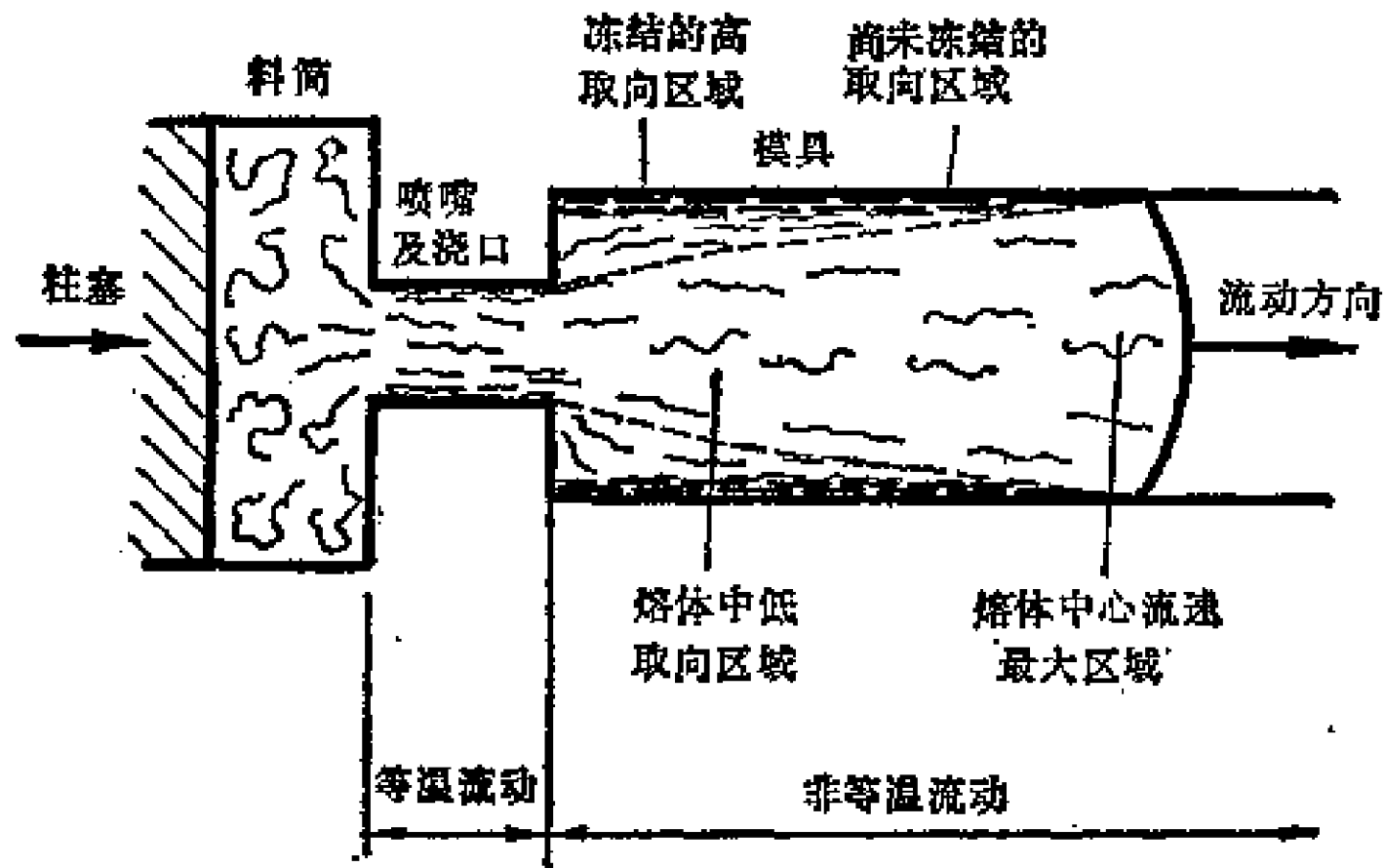
1. The viscosity and elasticity ratios, dynamic interfacial tension coefficient, critical capillarity number, composition, and flow field type and its intensity all influence the drop size.
2. In Newtonian liquid systems subjected to a simple shear field, drop breakup is the easiest when the viscosity ratio falls within the range $0.3 < \lambda < 1.5$. Drops with $\lambda \geq 3.8$ cannot be broken in shear.
3. Droplet breakup is easier in the elongation flow field than in the shear flow field. The relative efficiency of the elongational field dramatically increases for $\lambda \geq 3$.
4. Drop deformation and breakup in viscoelastic systems seems to be more difficult than that observed for the Newtonian ones.
5. When the concentration of the minor phase exceeds a critical value, $\phi_d > \phi_c \approx 0.005$, the effect of coalescence must be taken into account.
6. The theoretical predictions of droplet deformation and breakup are limited to infinitely diluted, monodisperse Newtonian systems.

8.3 聚合物及其填充体系的剪切取向

聚合物在加工过程中在管道型腔中的流动都是剪切流动，在剪切流动中速度梯度的作用下，卷曲状长链分子逐渐沿流动方向舒展伸直和取向。

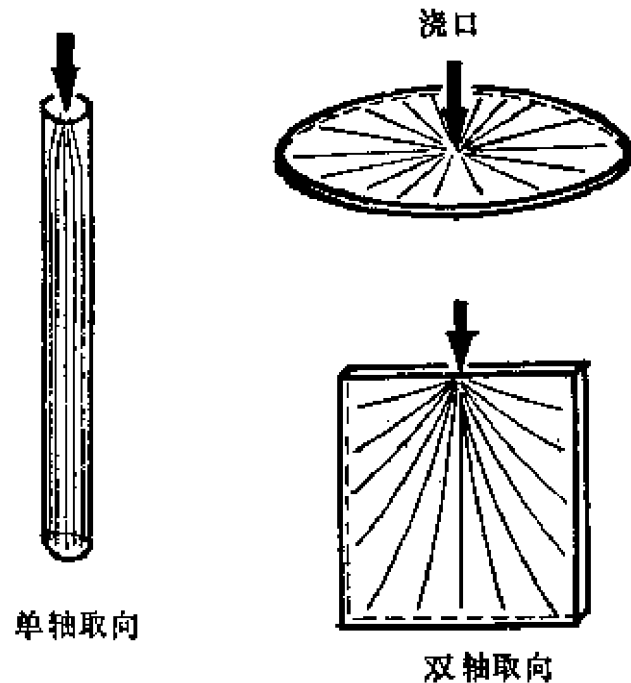
熔体温度高，分子热运动剧烈，因此在大分子流动取向的同时必然存在着解取向。

A. 聚合物在管道中和模具中的流动取向

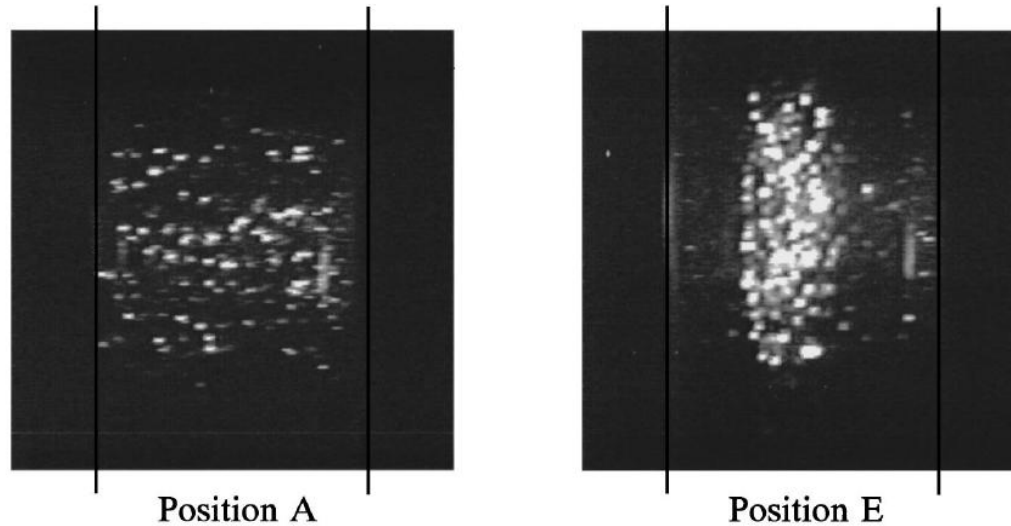


B. 流动取向可以是单轴或双轴的

取决于制件的结构形状、尺寸和熔体在其中的流动，截面积恒定，单轴取向，截面积变化，双轴取向或在更好的方向上取向。



C. 剪切导致悬浮粒子的迁移



Laser images corresponding to positions (A) near the entrance and (E) near the exit of the pipe for the same flow conditions. Accumulation of particles near the pipe axis at (E) shows significant radial migration during flow in the pipe.

Influence of shear deformation on carbon nanotube networks in polycarbonate melts: Interplay between build-up and destruction of agglomerates

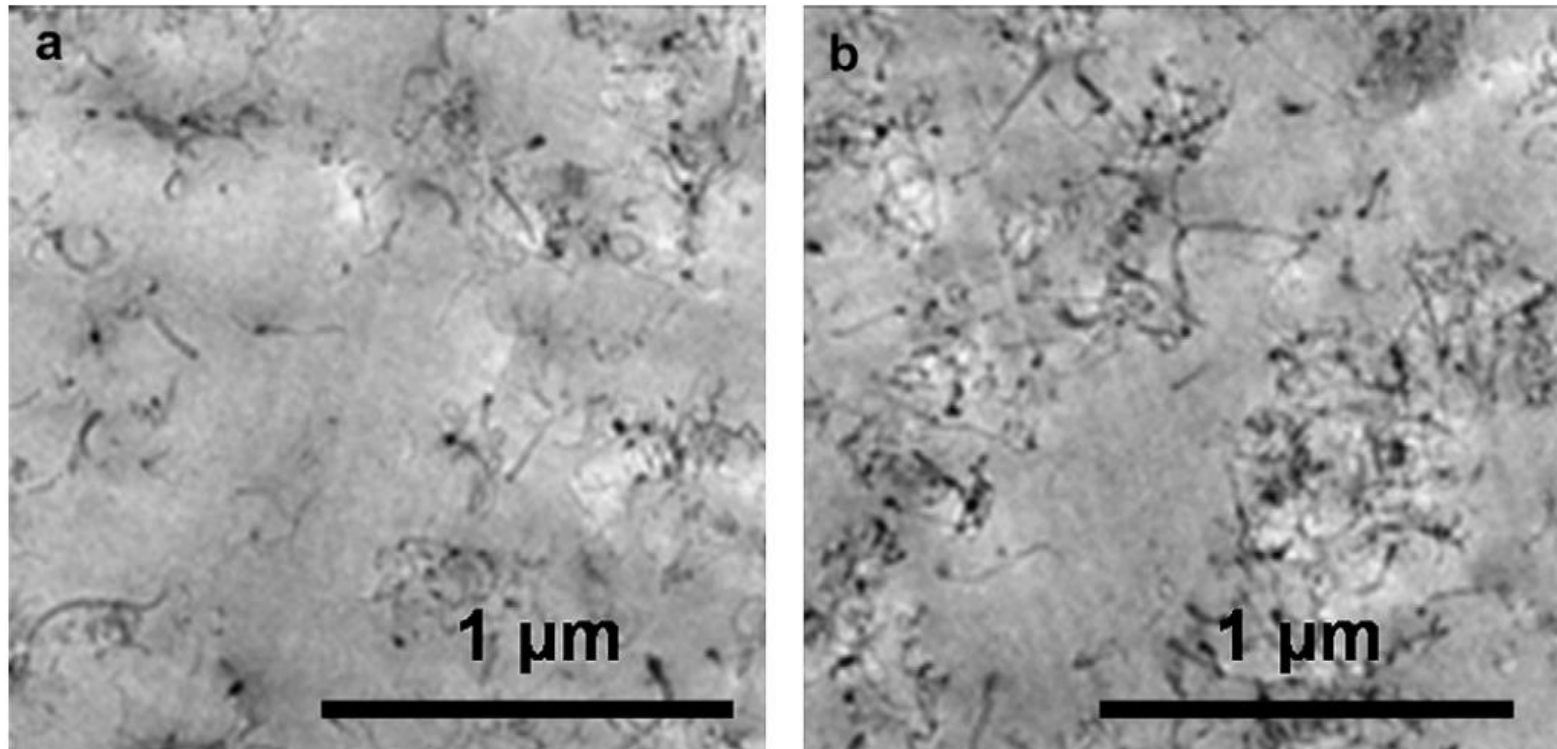


Fig. 1. TEM images of the as-produced PC/MWNT with 5 wt% MWNT (a) and after 1 h of annealing at 300 °C (b) [35].

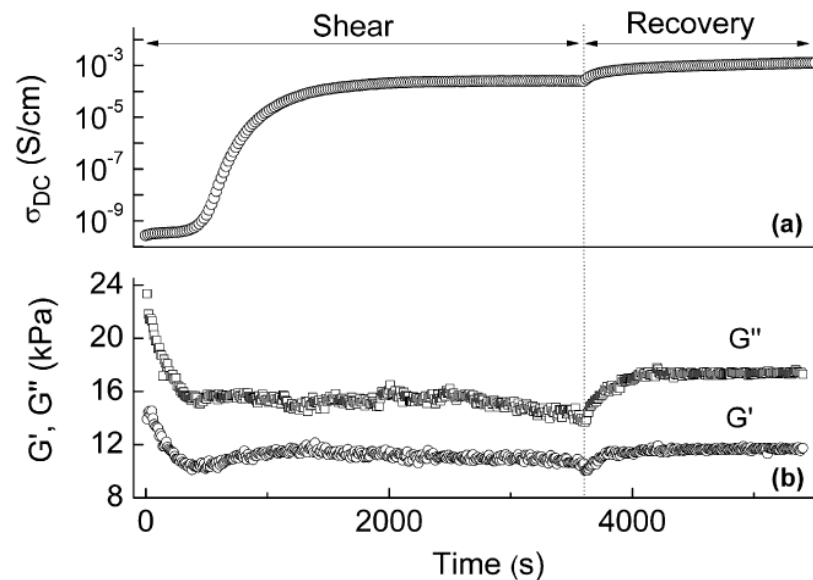


Fig. 6. Shear-induced insulator–conductor transition (a) during 1 h of steady shear (shear rate: 0.02 rad/s) and the quiescent recovery after shear and (b) simultaneously measured storage G' and loss modulus G'' during 1 h of shear and the quiescent recovery after shear (1 wt% MWNT).

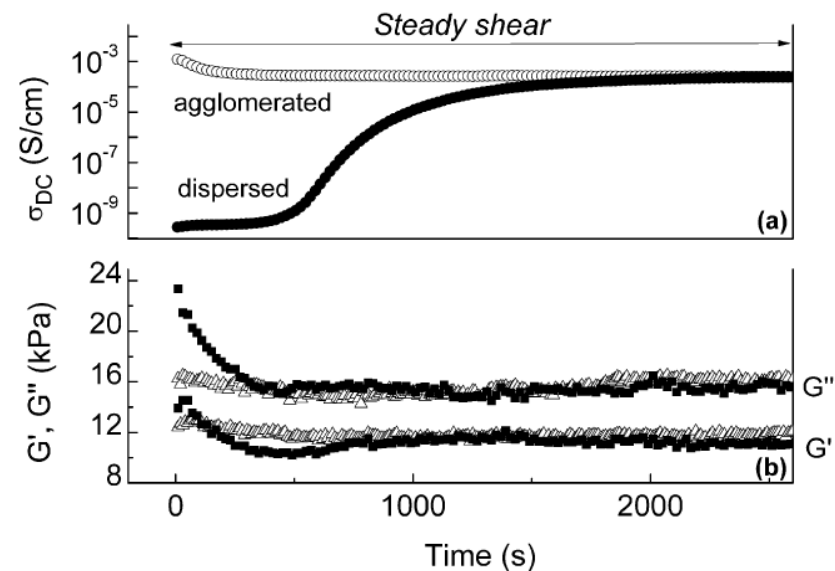


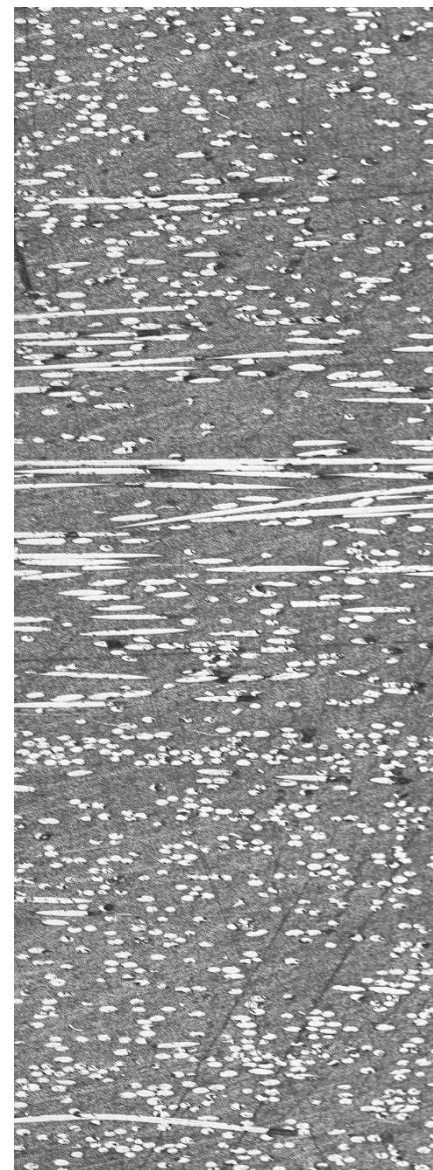
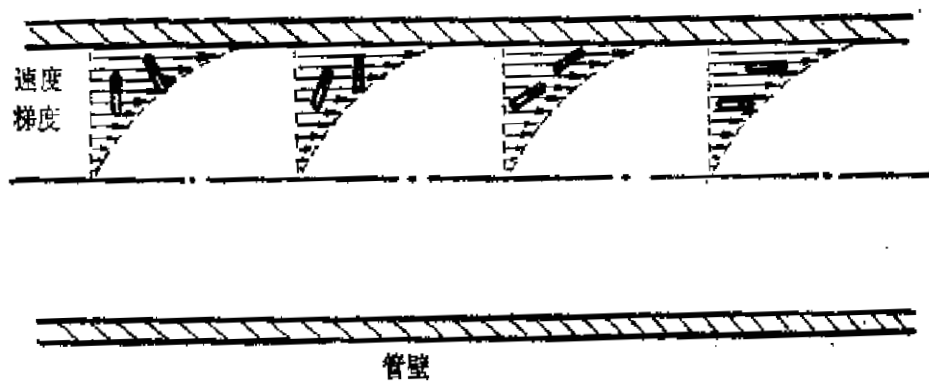
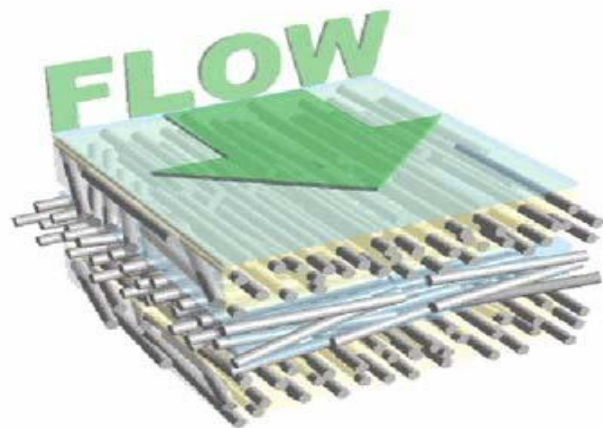
Fig. 7. Time dependence of the DC conductivity (a) during 1 h of steady shear (shear rate: 0.02 rad/s) and the simultaneously measured components of the complex shear modulus G' and G'' (b) for two samples (1 wt% MWNT) with initially 'dispersed' (solid symbols) and 'agglomerated' nanotubes (open symbols).

Shear-induced destruction and formation of conductive and mechanical filler networks formed by multi-wall carbon nanotubes in polycarbonate melts were investigated by simultaneous time-resolved measurements of electrical conductivity and rheological properties under steady shear and in the quiescent melt. The steady shear experiments were performed at shear rates between 0.02 and 1 rad/s and for nanotube concentrations ranging from 0.5 to 1.5 wt%. The influence of thermo-mechanical history on the state of nanotube dispersion and agglomeration was studied in detail.

For melts with well-dispersed nanotubes a *shear-induced insulator–conductor* transition was observed, which is explained by the agglomeration of nanotubes under steady shear and the formation of an electrical conductive network of interconnected agglomerates. Simultaneously, a drastic decrease of the shear modulus ($G^* = G' + iG''$) during steady shear was observed, which can be related to a reduction of mechanical reinforcement due to agglomeration of dispersed nanotubes. These findings indicate a substantial difference in the nature of “electrical” and “mechanical” network and contradict earlier assumptions that steady (or transient) shear is always destructive for the conductive filler network in highly viscous polymer composites.

It was also shown that after a certain time of steady shear the filler network asymptotically reaches its steady state characterized by the constant electrical conductivity and shear modulus of the composite melt. Such asymptotic behaviour of composite properties was experimentally shown to be related to the interplay of the destructive and build-up effects of steady shear. For modelling of the electrical conductivity in presence of steady shear a kinetic equation was proposed for filler agglomeration with shear-dependent destruction and build-up terms. This equation was coupled to the generalized effective medium (GEM) approximation for insulator–conductor transition.

D. 流动对填料取向的影响



8.4 聚合物的拉伸取向

8.4.1 非晶态聚合物的拉伸取向

1. T_g 附近的拉伸取向

当 $\sigma < \text{屈服应力} \sigma_y$ 时，只产生弹性形变

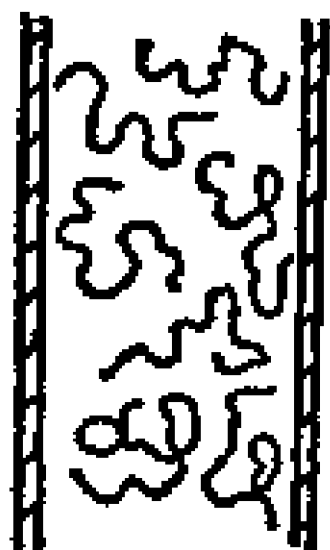
当 $\sigma > \text{屈服应力} \sigma_y$ 时，可进行塑性拉伸

2. 在 $T_g \sim T_f$ 间拉伸

T 足够高时， σ_y 几乎不显著，不大的外力就可使聚合物产生连续的均匀的塑性形变，并可获得较高稳定的取向结构，这时材料的形变是均匀的拉伸过程。

3. 粘流温度 T_f 以上的拉伸

分子活动能力提高，大分子易解缠，滑移和取向，但同时解取向速度也提高了，因此有效取向程度低，可迅速冷却保持取向度，熔融纺丝的取向就是粘流取向。



未取向

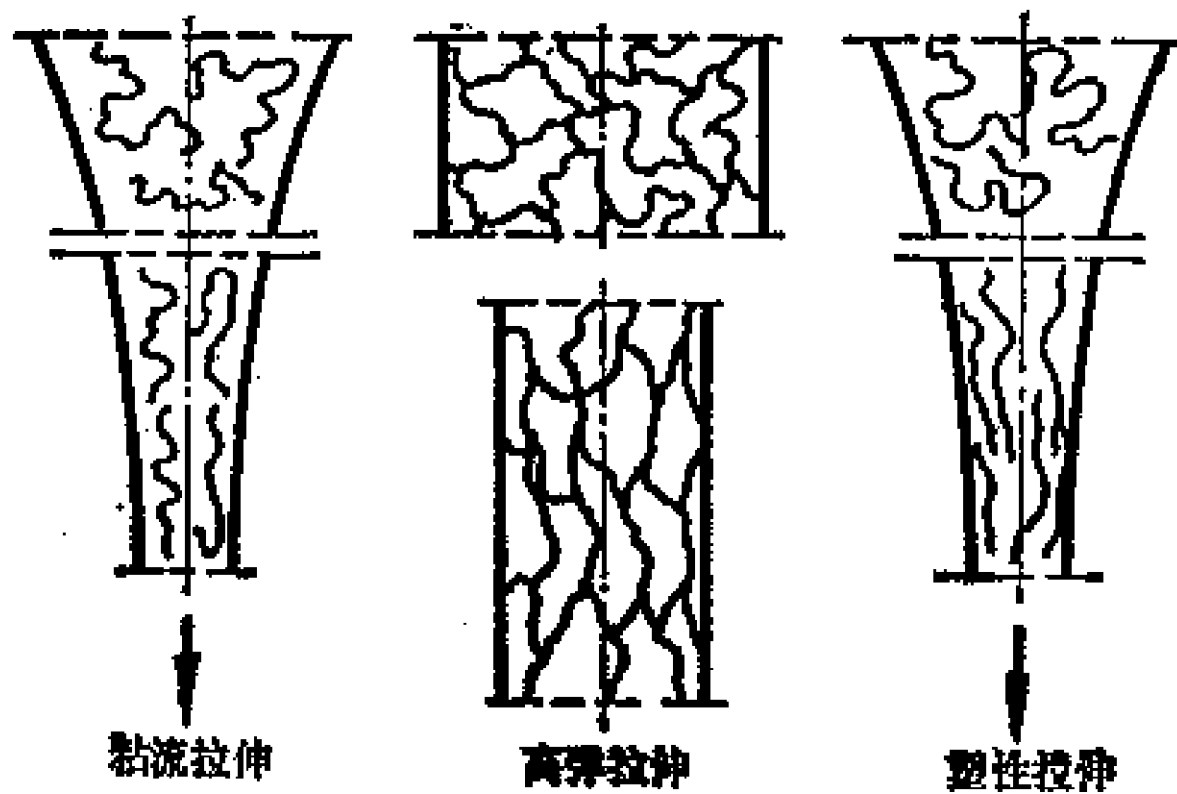


链段取向



大分子取向

非晶态聚合物的取向示意



聚合物三种拉伸机理的示意

8.4.2 晶态聚合物的拉伸取向



非晶聚合物的拉伸取向



结晶聚合物的拉伸取向：晶体倾斜、破裂、熔化、取向、再结晶

取向后，沿分子链方向上，拉伸强度、拉伸模量增大

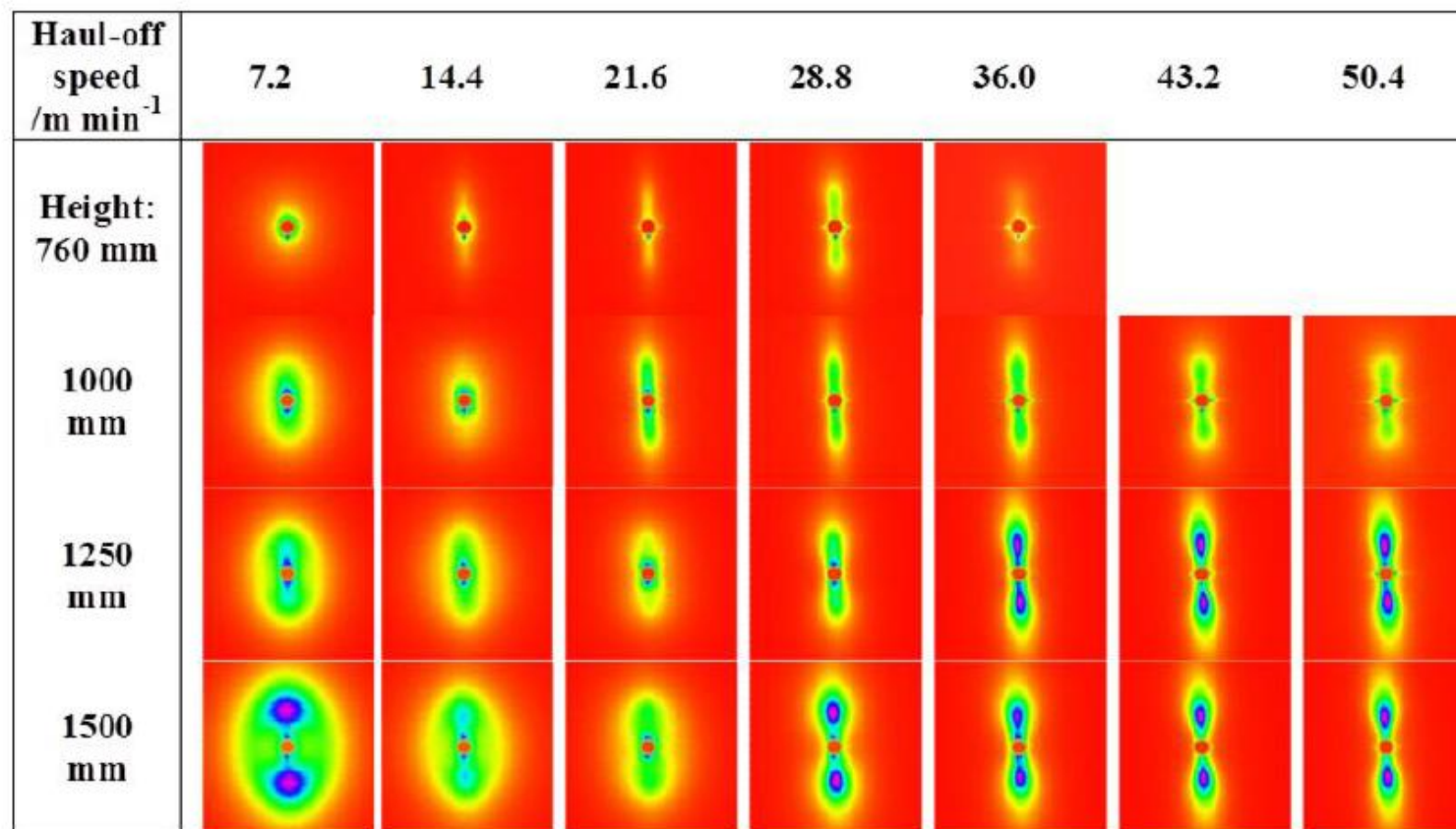


Figure 3. Matrix of 2D SAXS patterns of LDPE tape at various heights and haul-off speeds with a mass flow rate of 5.06 g/min and die temperature 150 °C.

Heeley, E. L.; Gough, T.; Hughes, D. J.; Bras, W.; Rieger, J. and Ryan, A. J. (2013). Effect of processing parameters on the morphology development during extrusion of polyethylene tape: an in-line Small-Angle X-ray Scattering (SAXS) study. *Polymer*, 54(24) pp. 6580–6588.

A. 温度T的影响

$T \uparrow \rightarrow \eta_a \downarrow \rightarrow$ 有利于取向；

$T \uparrow \rightarrow$ 分子热运动 $\uparrow \rightarrow$ 松弛时间 $\downarrow \rightarrow$ 有利于解取向。

取向过程是个平衡过程。

取决因素：

a. $T_p - T_s$ 温区宽，易解取向；

b. 松弛时间短，易解取向；

c. $T_p - T_s$ 的冷却速度；冷却速度与冷却温度 T_c 有关，还取决于其他参数：比热 $\uparrow \rightarrow$ 冷却速度 $\downarrow$ 有利于解取向。

B 应力的影响

应力提高，取向度提高，但应力与温度分不开。

a. $T_g \sim T_f$ 区间，小应力就能使大分子取向。

b. 室温，冷拉，需较大应力，

c. 当 σ 、 ε 不变时，拉伸中的热效应使材料 T 升高，

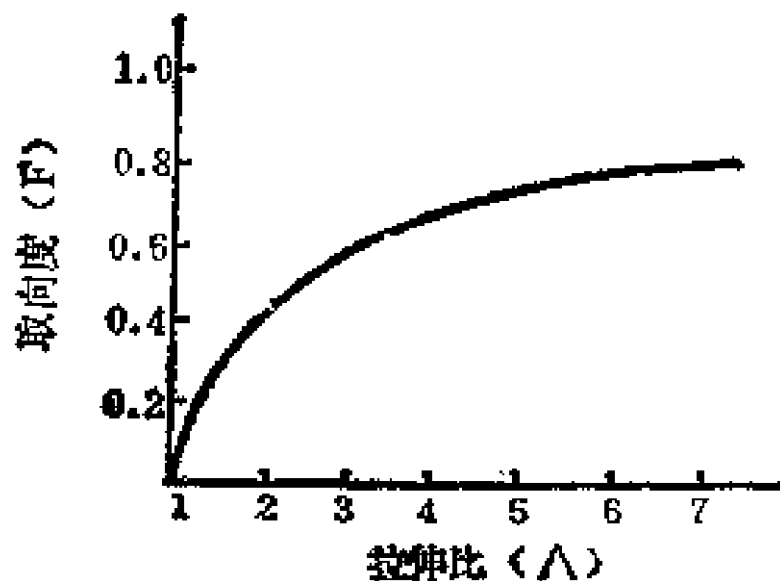
造成粗细、厚薄、取向的不均匀，性能变差。

因此，等温拉伸，可获得稳定的取向材料。

C 拉伸比的影响

在一定温度下材料在屈服应力作用下被拉伸的倍数称为自然拉伸比 $\Lambda=L/L_0$

被拉伸材料的取向程度随拉伸比而增大



8.4.2 聚合物结构和低分子物的影响

A、聚合物结构

1. 柔性好，易取向和解取向，必须结晶才能得到取向结构。如PE、PP、POM。
2. 刚性强，不易取向和解取向，但一旦取向，较稳定。
3. 分子量低，易取向和解取向，必须结晶冻结取向结构。
4. 分子量高，不易取向和解取向，形成较稳定的取向结构。
5. 分子间作用力，有利于取向结构的稳定。
6. 结晶，不易取向和解取向，取向结构稳定。如PA、PET。

B、低分子物

溶剂、增塑剂使 T_g 、 T_f 下降，松弛时间缩短，形变加速，易取向和解取向，取向后如果使溶剂挥发或凝胶，取向可稳定下来。

8.5 流动对结晶的影响

剪切力、拉伸力的作用使分子取向，形成有序排列，结晶速率和成核密度提高

题5：你推测一下，多大的剪切速率可以使得聚丙烯熔体（140摄氏度，重均分子量44万，数均分子量9.6万）分子链发生取向？

如何估算分子链的最长松弛时间？

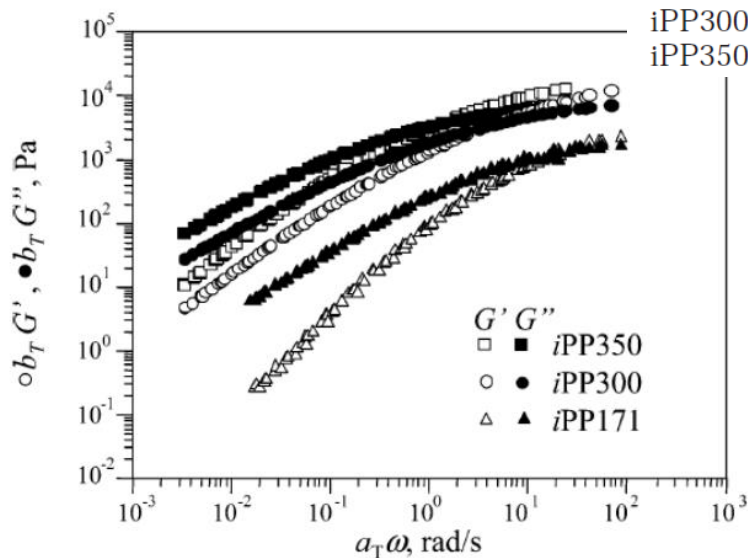


Figure 1. Melt rheology of *iPP* samples. Storage modulus G' and loss modulus G'' of *iPP* samples shifted to 145 °C.

sample	M_w (kg/mol)	M_n (kg/mol)	M_w/M_n
iPP171	171	40.4	4.2
iPP300	300	51.8	5.8
iPP350	350	85.8	4.1

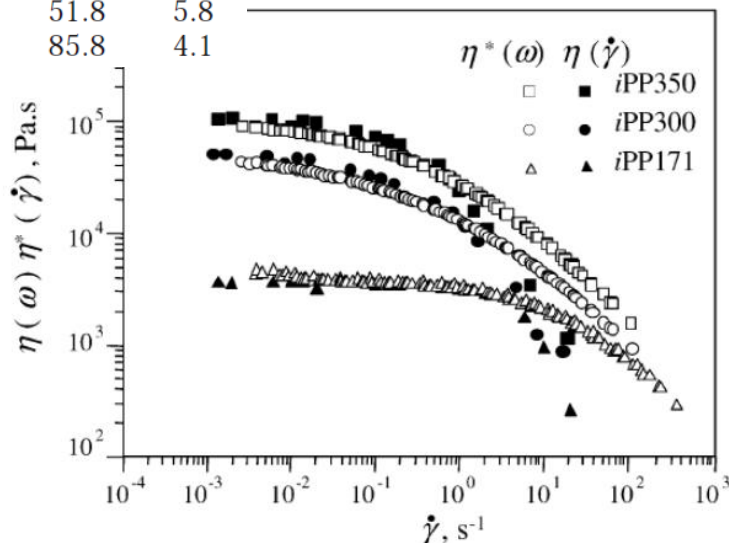
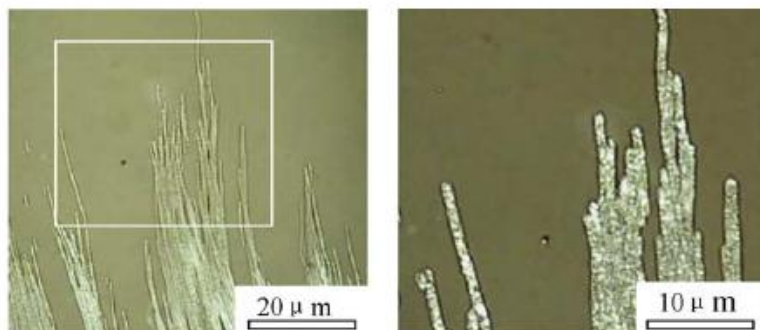


Figure 2. Shear viscosity as a function of shear rate at 145 °C along with dynamic viscosity curves from time-temperature superposition shifted to 145 °C. Crossover from zero shear

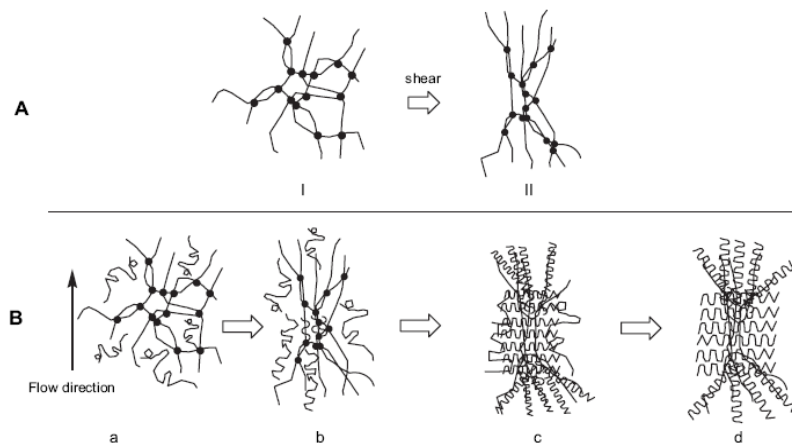
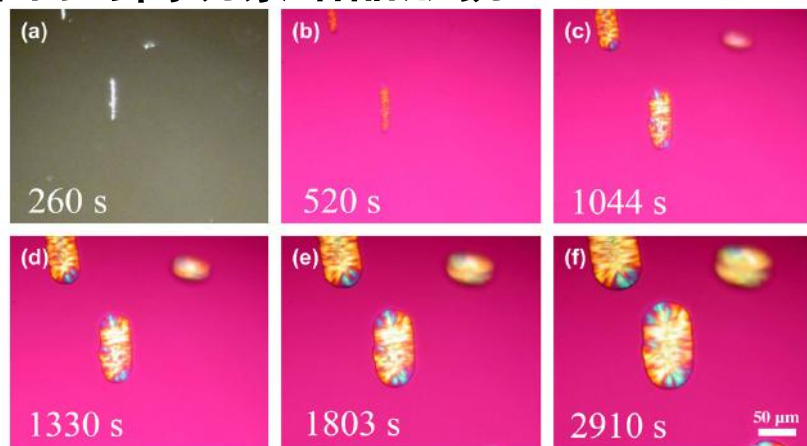
A. 熔体预剪切对结晶形貌的影响



(a) 142 s

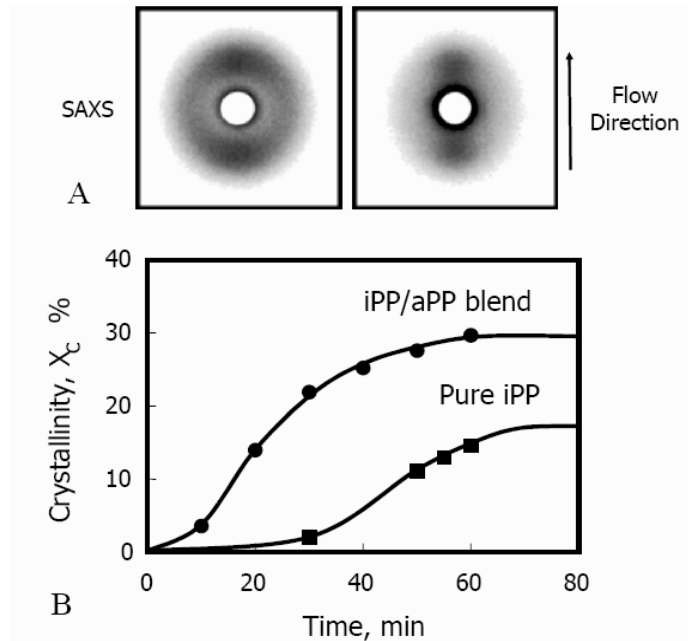
(b) 419 s

PP熔体140°C剪切5 s，剪切速率 0.5 s^{-1} ，剪切停止后不同时间观察结晶形貌



剪切导致熔体缠结网络取向，取向结构结晶形成串晶形貌

PP熔体140 ° C剪切2 s，剪切速率 0.5 s^{-1} ，剪切停止后不同时间观察



aPP, $M_w = 670,000$ g/mol, PDI= 2.6 (5%)

iPP, $M_w = 127,000$ g/mol, PDI = 2.3 (5%)

等规PP和高分子量无定型aPP共混，证明是无定型网络取向而不是等规PP分子量取向结晶形成shish诱导的取向结晶

B. 剪切对结晶速率和成核密度的影响

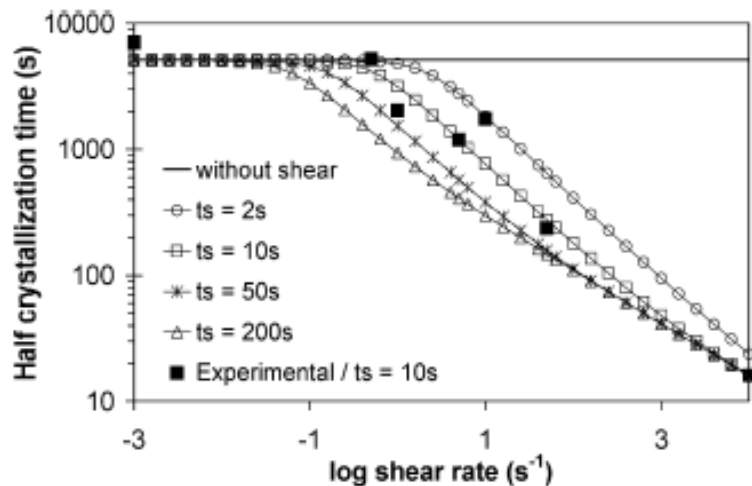


Fig. 14. $T_c = 140\text{ }^{\circ}\text{C}$ /experimental and calculated half crystallization time for different shearing times.

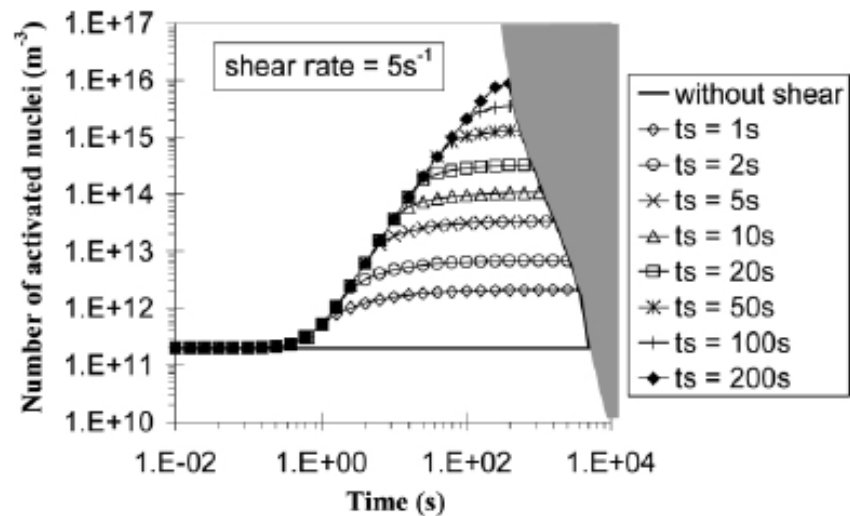


Fig. 13. $T_c = 140\text{ }^{\circ}\text{C}$ /calculated number of activated nuclei as a function of time for different shearing times.

存在剪切诱导结晶的最低剪切速率，高于该剪切速率才能观察到剪切诱导结晶(用韦森堡数解释)

在相同剪切速率下，总结晶速率和成核密度随剪切时间延长而变快。

在相同剪切时间情况下，总结晶速率随剪切速率升高而变快

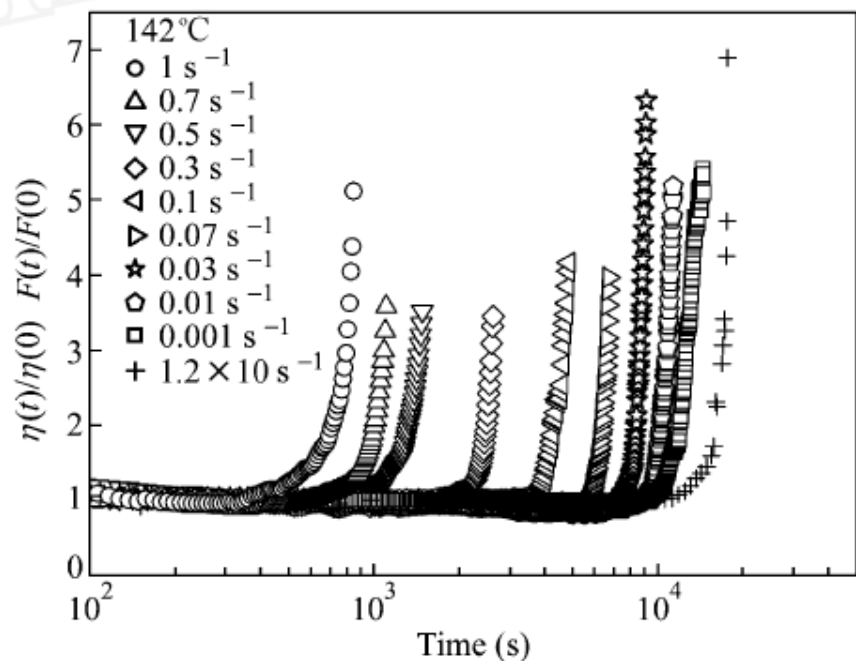


Fig. 2 Normalized viscosity curves at 142 °C, in the range of shear rate from $1.2 \times 10^{-4} \text{ s}^{-1}$ to 1.0 s^{-1}

流变测试粘度的升高
反映了结晶度的提高

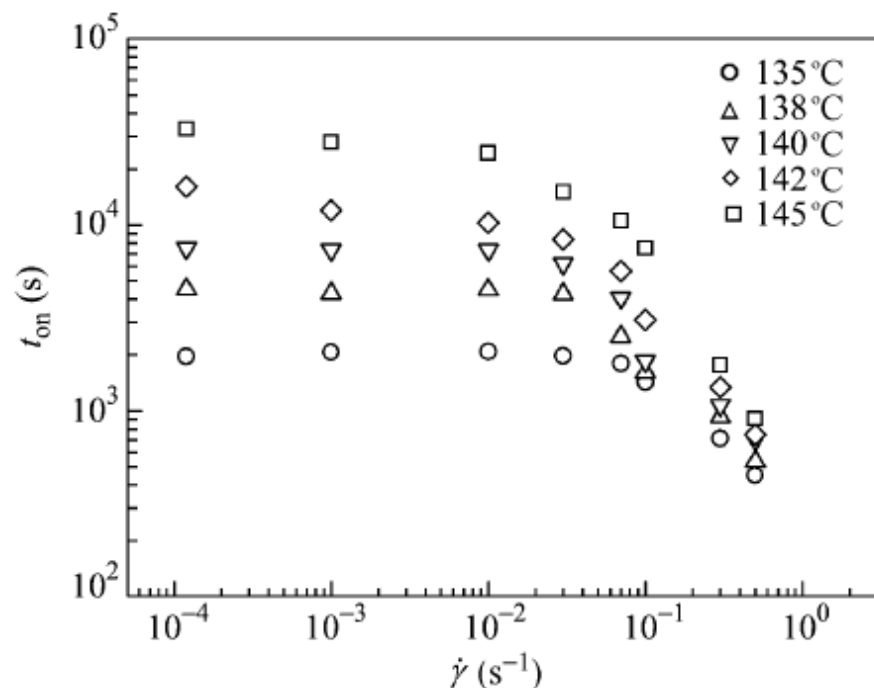


Fig. 3 Dependence of the onset time on the shear rate at various crystallization temperatures

结晶起步时间随剪切速率增加而
缩短, 反映了主成核速率的提高

陈青, 范毓润, 郑强。高分子学报, 2007, (5), 467-471.

C. 剪切诱导结晶存在临界剪切应变

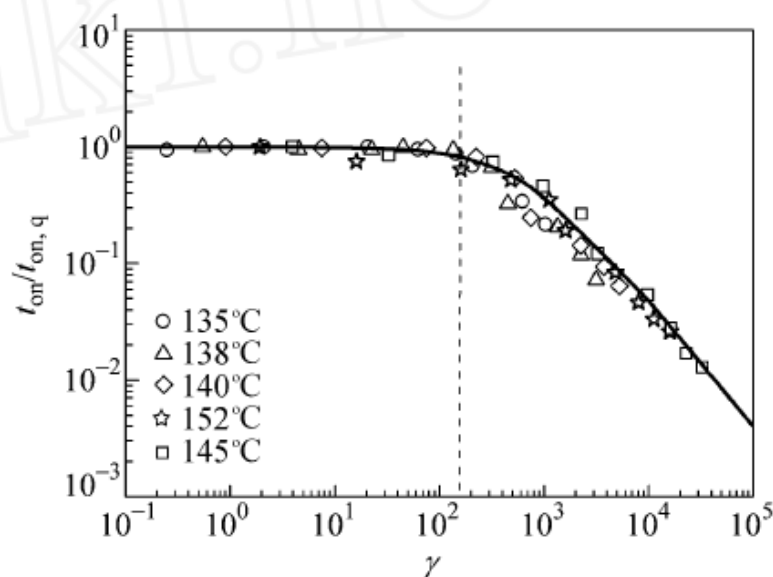


Fig. 5 The normalized onset times versus onset strain of *i*PP at different crystallization temperatures
Line is the predictions of Eq. (2) and the point is experimental data.

不同温度下结晶起步时间和应变作图, 获得一条主曲线
反映了主成核速率的提高取决于剪切应变

需要足够的剪切应变, 才能产生诱导结晶所需的最低取向度

Correlation of Material and Processing Time Scales with Structure Development in Isotactic Polypropylene Crystallization

Table 1. Characteristics of Samples Utilized To Study the Effects of Molar Mass and Shear on the Crystallization of *i*PP^a

sample	M_w (kg/mol)	M_n (kg/mol)	M_w/M_n	λ_i , s	λ_η , s	λ_η/λ_i
iPP171	171	40.4	4.2	5.50×10^{-2}	4.56×10^{-1}	8.20
iPP300	300	51.8	5.8	2.17×10^{-1}	1.54	7.10
iPP350	350	85.8	4.1	6.30×10^{-1}	4.76	7.94

^a Characteristic relaxation times of *i*PP samples at 145 °C; λ_i : crossover from terminal behavior to entanglement. λ_η : crossover in steady-state shear viscosity curve from a plateau in η to the shear-thinning region.

average over the slow modes of the spectrum. For lack of a more specific time constant, rheologists often define a representative longest relaxation time with the intersect frequency ω_i at which dynamic moduli G' and G'' intersect (subscript “i” for intersect) in mechanical spectroscopy or by the onset of shear thinning at a characteristic shear rate $\dot{\gamma}_c$ in a steady shear experiment:

$$\lambda_i = \frac{1}{\omega_i} \quad \text{or} \quad \lambda_\eta = \frac{1}{|\dot{\gamma}_c|} \quad (1)$$

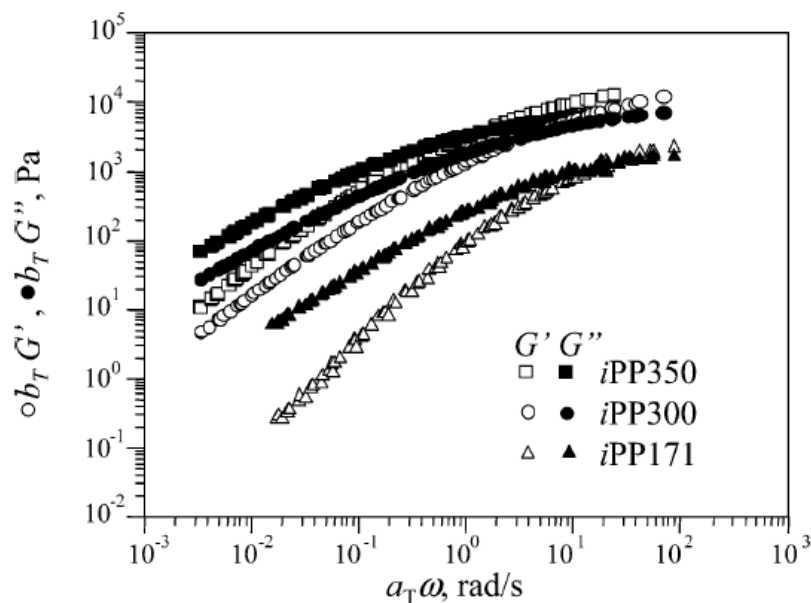


Figure 1. Melt rheology of *i*PP samples. Storage modulus G' and loss modulus G'' of *i*PP samples shifted to 145 °C.

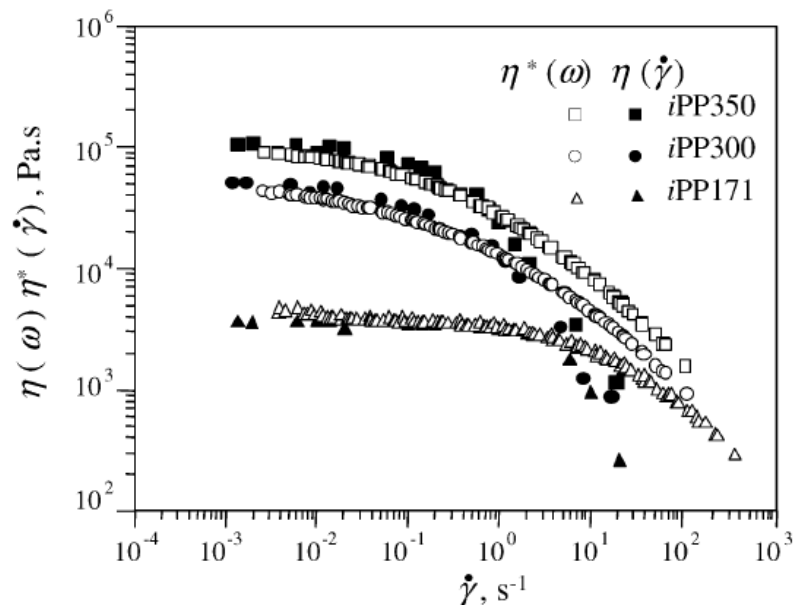


Figure 2. Shear viscosity as a function of shear rate at 145 °C along with dynamic viscosity curves from time–temperature superposition shifted to 145 °C. Crossover from zero shear

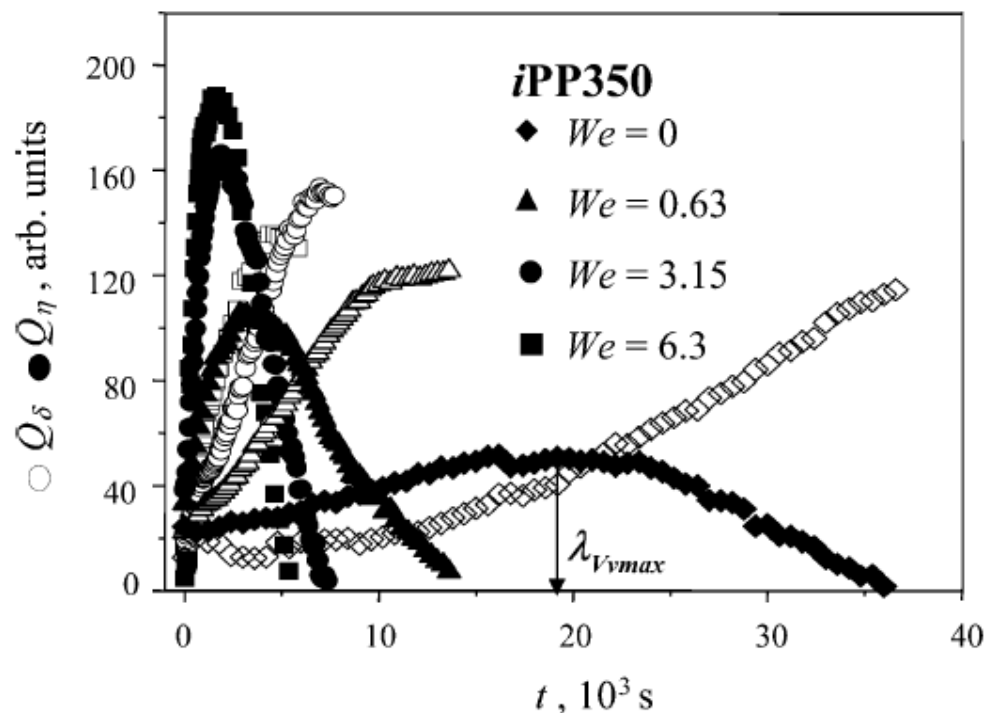


Figure 5. Time evolution of SALS invariants Q_η and Q_δ of *iPP* samples, quiescent and sheared ($\dot{\gamma} = 1, 5$, and 10 s^{-1} , with $t_s = 600, 120$, and 60 , respectively), at increasing We values. λ_{Vvmax} is crystallization time corresponding to instant at maximum density fluctuations. Density fluctuation invariant Q_η in full symbols and orientation fluctuation invariant Q_δ in open symbols.

$$Q_\delta = \int I_{Hv} q^2 dq = \frac{2\pi^2}{15} \left(\frac{\omega}{c} \right)^4 \langle \delta_n^2 \rangle$$

$$Q_\eta = \int \left(I_{Vv} - \frac{4}{3} I_{Hv} \right) q^2 dq = 2\pi^2 \left(\frac{\omega}{c} \right)^4 \langle \eta^2 \rangle$$

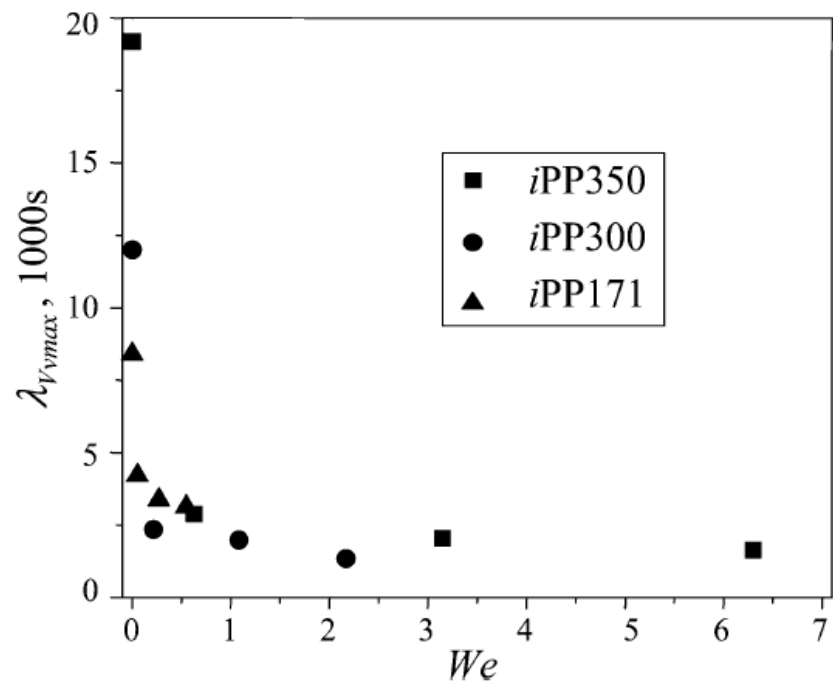


Figure 6. Characteristic crystallization time λ_{vmax} vs Weissenberg number for all molar mass samples.

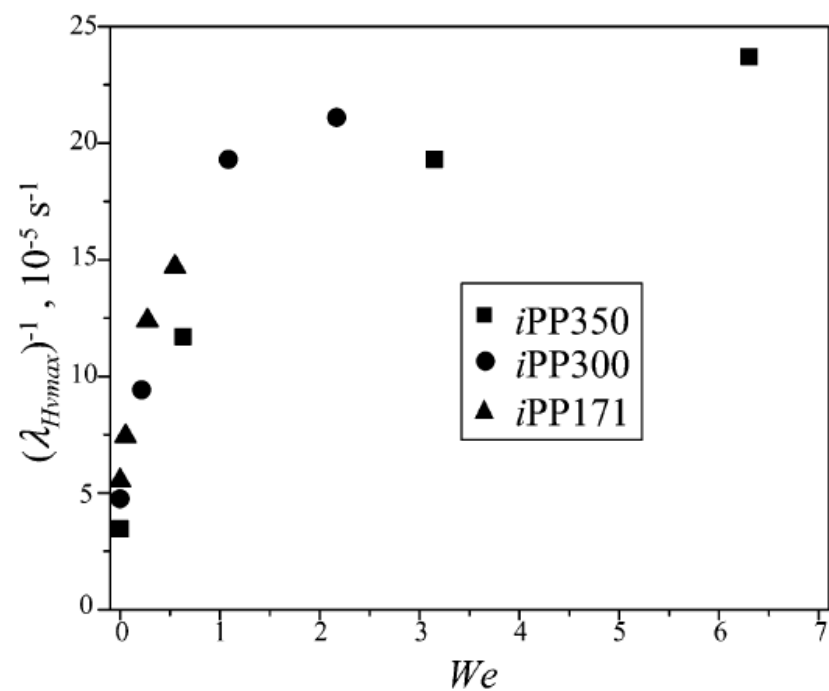


Figure 8. Reduced orientation fluctuation slopes of all molar mass samples at increasing Weissenberg values.

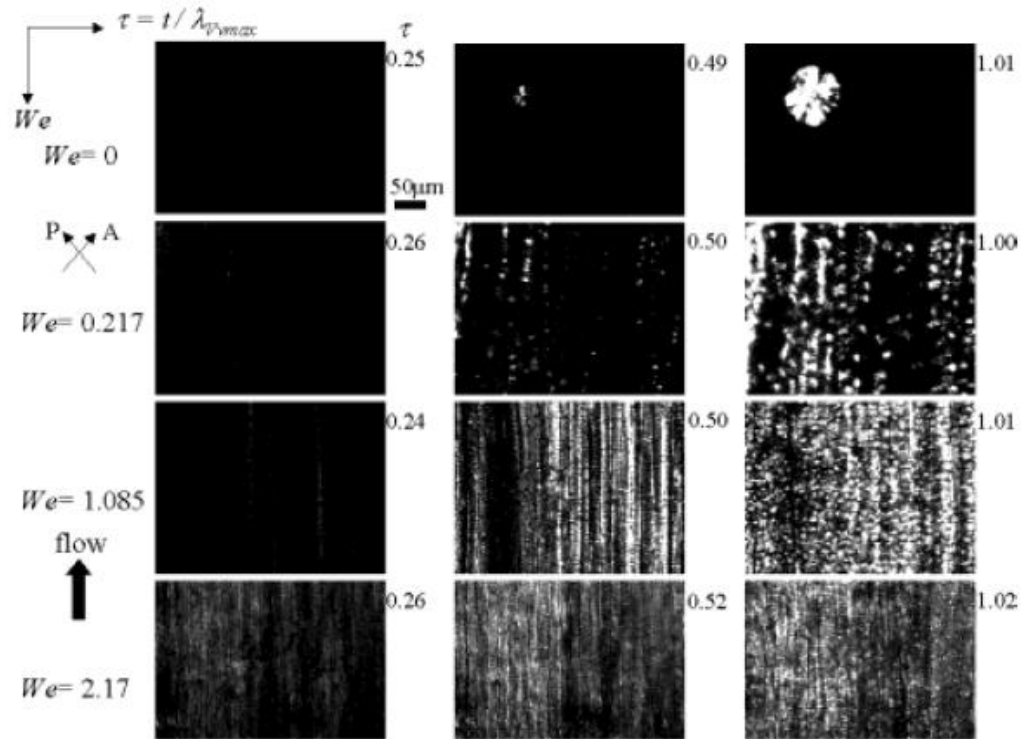
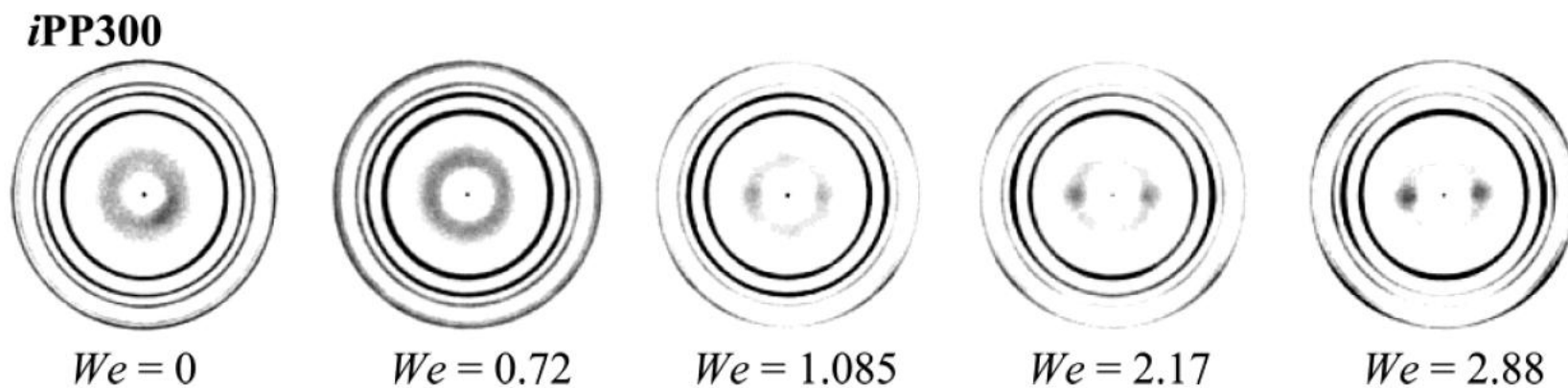


Figure 10. Optical micrographs showing evolution of quiescent and shear-induced crystallization of *i*PP300 at increasing We values. Micrographs shown at increasing normalized time scale $\tau = t/\lambda_{Vmax}$ for comparison.



D. 流动诱导结晶小结

1. 存在最低临界剪切速率（剪切应变），大于此剪切速率后才能观察到流动加速结晶。

Weissenberg数 We

2. 微观原因主要是流动诱导聚合物网络的取向。取向结晶结构的形成随剪切时间增加而延长。
3. 熔体取向结构在停止剪切后随时间延长而逐渐松弛。
4. 剪切和拉伸导致聚合物结晶成核速率增加。其机理包括：剪切取向使平衡熔点升高；外场作用下成核能垒下降。
5. 熔体取向松弛和结晶的竞争导致最终的结晶结构取向度。

需要掌握的公式

1. 粘度的幂律公式和普适粘度曲线
2. 粘度的自由体积理论和微观速率理论
3. 管中流体（牛顿与非牛顿）的流动
4. 毛细管流变仪和三种旋转流变仪测定粘度的公式
5. 线性粘弹性的Maxwell 模型
6. 松弛时间、零切粘度的公式
7. 动态模量和稳态剪切的对应关系
8. 两个无量纲数：

Deborah number, Weissenberg number